

**Making Room for Theory in DIF and Measurement Invariance Analysis with
Covariate Effects on Groups of Items**

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Abstract

We propose and study an extension of moderated nonlinear factor analysis that models covariate effects—that is, differential item functioning (DIF)—at the level of item groups rather than, or in addition to, individual items. This allows for a more theory-driven approach to DIF and measurement invariance testing in which researchers can form and test hypotheses about *why* items differ in how they function across an arbitrary number of categorical or continuous covariates. As a motivating example, we demonstrate gender effects on a developmental measure where scores that assume invariance overstate boys’ cognitive development and understate their fine and gross motor development, with opposite implications for girls. Simulation demonstrates the empirical identifiability and predictable statistical performance of the proposed model, and additional empirical examples demonstrate the approach’s substantial flexibility.

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Introduction

Differential item functioning and measurement invariance are two sides of the same coin, and on both sides there is dissatisfaction. Differential item functioning (DIF; Angoff & Ford, 1973; Holland & Thayer, 1986) analysis is primarily concerned with the identification of specific items that do not function equivalently across relevant background variables. On the other side, measurement invariance (MI; Meredith, 1993) analysis is a matter of determining the types of parameters in a measurement model that are (or are not) equivalent across covariates so that structural relations among latent variables can be interpreted as general to the latent variables rather than specific to a subset of items. Fundamentally, they are both about the role of covariates in what happens when a person of a given level on a latent variable interacts with an item with a given set of parameters defining how it responds to that latent variable (Thissen, 2025). DIF is, historically, part and parcel with the analysis of bias in the items used on educational tests (Angoff, 1982; Lord, 1980). In this context, DIF is a statistical phenomenon that the researcher seeks to substantively interpret as item bias. Dissatisfaction in contemporary DIF analysis comes from increasing recognition that the items that exhibit DIF as a statistical phenomenon do not necessarily do so for construct-irrelevant reasons (AERA et al., 2014), meaning that one can routinely find DIF without bias (Ackerman & Ma, 2024; Camilli, 1993). More formally, DIF analysis alone generally cannot rule out alternative causal explanations for why a given item might not function equivalently for respondents who differ with respect to some covariates, limiting what one can learn from a DIF analysis (Suk & Lyu, 2026).

On the MI side of the literature, researchers have come to essentially the same conclusion, but in the context of a different measurement tradition (Jöreskog, 1971; Sörbom, 1974; Thissen, 2025): violations of invariance on their own do not necessarily mean the latent variable(s) under analysis cannot be meaningfully compared across

covariates (Robitzsch & Lüdtke, 2023). Moreover, through analysis of *which items* exhibit noninvariance with respect to *which covariates*, MI analysis can be made into a substantively informative method of understanding how phenomena manifest across individuals of different backgrounds (rather than simply a statistical hoop to jump through to justify the use of a larger model; Rohrer & Paulewicz, 2025; Sterner et al., 2024, 2026). We read these two literatures as converging on the idea that whether one approaches differences in how people of different backgrounds respond to items through the lens of DIF or MI, historically popular methods leave information on the table and are of limited use for the task of learning about the psychometric properties of an instrument.

One reason for this state of affairs may be that in traditional item response theory (IRT) and factor analysis models for DIF and MI, there is almost no room for information about the items themselves. These models are made up of parameters describing statistical relations between respondents' standing on (a) latent variable(s) and observed patterns in item responses. Modeling DIF or MI is a matter of freely estimating differences in these parameters with respect to covariates, either on an item-by-item basis as in DIF analysis or on a parameter-by-parameter basis as in MI analysis. Either way, what is missing (and that we believe may be useful in many of these analyses) is a means to capture researchers' hypotheses about *why* items may exhibit DIF in terms of the content of the items. The aim of this paper is to outline methods by which this can be accomplished with respect to any of the parameters of a general measurement model. We propose an extension to the increasingly popular moderated nonlinear factor analysis (MNLFA; Bauer, 2017; Bauer & Hussong, 2009) model that decomposes DIF effects into effects on groups of items, and additional item-specific effects above and beyond the item group-level.

The Role of Theory in Understanding Differential Item Functioning and Measurement Noninvariance

This paper introduces an approach to modeling DIF (or measurement noninvariance) that allows for theory-driven effects on groups of items, as well as additional

idiosyncratic DIF effects on individual items above and beyond these groups of items. We present these models as extensions of MNLFA models because of this framework’s emphasis on generality with respect to both covariates and item types in DIF and MI analysis.

While multiple-group (Muthén & Lehman, 1985) and multiple indicators, multiple causes (MIMIC; Jöreskog & Goldberger, 1975; Woods & Grimm, 2011) models can capture certain types of DIF, MNLFA subsumes both approaches to enable the modeling of DIF with respect to any parameter in a measurement model by an arbitrary number of covariates, categorical or continuous (Bauer, 2023).

A reasonable question is, what do we mean by “theory-driven?” Many psychological instruments, even (or especially) those intended to be interpreted via a unidimensional score, contain explicit or implicit “groups” of items that have *something in common*. What they have in common depends on the application, but consider the following examples (the first two being from empirical reanalyses presented later in this paper):

1. A measure of child development contains some items that measure cognitive development and some items that measure motor development (Weber et al., 2019).
2. A measure of vocabulary development (naturally!) contains a mix of adjectives, nouns, and verbs (Bohn et al., 2025).
3. A significant subset of the items on a large-scale international math assessment are about fractions, though items also draw on many other domains (Braeken et al., 2026).

In each case, the items fall somewhat short of the psychometric ideal of items that are completely interchangeable reflectors of the underlying construct with nothing in common except the construct itself; in our view this is the reality of many, if not most, educational and psychological scales and not an idiosyncrasy of these specific instruments. After all, the first example comes from a paper developing a measure that has since been endorsed as a universal metric for child development by the World Health Organization

(McCray et al., 2023), while the third is from one of the most heavily developed and studied psychometric instruments in the world, the Trends in International Math and Science Study (TIMSS). More broadly, truly unidimensional constructs are very rare (Bonifay et al., 2015), with even guides on best practices in measurement generally acknowledging that scale construction is a balance between statistical ideals and the realities of achieving sufficient content coverage (e.g., Bandalos, 2018; Clifton, 2020).

The fact that items frequently appear to form “groups” has direct implications for DIF/MI analysis. The basic premise of DIF in a parametric psychometric modeling framework (i.e., IRT or factor analysis) is that the parameters defining the item response function differ with respect to one or more covariates (Angoff, 1982; Lord, 1980); MI is concerned with the same issue, just with a focus on a specific parameter across items instead of a specific item with respect to its parameters (Meredith, 1964, 1993; Thissen, 2025). Over the last fifty-plus years an immense amount of research has gone into methods for finding items that exhibit DIF/noninvariance. Increasingly, emphasis in recent years has been on methods that automate this process (e.g., Belzak & Bauer, 2020; Brandt et al., 2025; S. M. Chen et al., 2022; Y. Chen et al., 2023; Ouyang et al., 2026; Robitzsch, 2024; Schauburger & Mair, 2020; Tutz & Schauburger, 2015). This is valuable when the researcher knows little about their items, but in a great deal of psychological research, this is simply not the case: for better or worse, psychological researchers frequently devise and administer their own measures (Cortina et al., 2020; Flake & Fried, 2020; Hussey et al., 2025). Even when using preexisting instruments, researchers are typically experts in the area they are studying, suggesting that they can engage closely with the content of their measures. In other words, it should be straightforward to many psychological researchers to identify items that form groups in the context of a given DIF analysis if such groups are theoretically relevant.

Recently, researchers have observed general neglect of MI in psychological research (Maassen et al., 2025). Others have subsequently speculated that, as noted above, this may

in part be because MI analysis essentially serves as a hoop to jump through before the “real” analysis can begin. The idea is that applied psychological researchers have nothing to gain from assessing MI because the only possible outcomes of doing so are (1) doing what the researcher was going to do anyway, or (2) not doing it because the measure failed invariance testing. This has been critiqued as leaving a great deal of theoretically-interesting information on the table (Rohrer & Paulewicz, 2025; Sterner et al., 2024, 2026), a perspective that informs the present paper. As noted above, classic and recent work in DIF has reached a similar conclusion (e.g., Ackerman & Ma, 2024; Camilli, 1993; Suk & Lyu, 2026), that DIF analysis based solely on empirical criteria internal to the instrument under investigation can be misleading because DIF can be caused not only by biased items, but also by legitimate differences in the construct that operate on subsets of items with respect to covariates and inform our understanding of how the construct operates in nonequivalent populations. Using theory to identify item groups—and assessing DIF/MI with respect to these item groups—changes the nature of the analysis; instead of focusing on finding and eliminating “bad” items (as in DIF analysis) or checking a box (as in MI), the focus on groups of items can help researchers probe the limits of interpretation of scores on an instrument. For example, consider DIF by sex of the child in the case of child development cited above. If, as we show below, there is evidence that there is DIF that operates on the cognitive vs. fine and gross motor items, this means that scores for boys vs. girls may likely over- or under-state girls’ vs. boys’ development in these domains. It is also important to recognize that the groupings that are relevant for one analysis may not be relevant for another; such is the nature of a theory-driven approach to DIF/MI. For instance, Braeken et al. (2026)’s focus on fractions vs. non-fractions items in response to a specific concern about relative performance on fractions items in Nordic countries, but that does not mean that there is no other way to group the items on TIMSS; appropriate groupings will always depend on context and the research question at hand.

We are far from the first researchers to consider item characteristics in psychometric

modeling, but we see our contribution as unique relative to existing approaches because it is situated entirely in the fixed-items measurement modeling paradigm and theory-driven, but also flexible with respect to item covariates. Next, we briefly review these to motivate the present work.

Previous Approaches to Accounting for Item Characteristics in Psychometric Models

Explanatory Item Response Models. The most common framework for including item characteristics in IRT-based models is the Explanatory Item Response Model (EIRM; De Boeck & Wilson, 2004). In EIRM, both person and item parameters can be decomposed into sums of person-side and item-side covariates. This decomposition aims to provide explanations for observed differences in the means of person ability distributions or differences in item difficulties. In this way, EIRM generalizes previous covariate-adjusted item response models, including the Linear Logistic Test Model (LLTM; Embretson, 1999; Fischer, 1973) and latent-regression based IRT models (Christensen, 2006). Subsequent work has relaxed the need for these decompositions to be linear combinations of covariates (Cho et al., 2024), but a key observation about EIRM is that it is typically set within the framework of generalized linear mixed models and, as such, allows for model parameters to be conceptualized as either fixed or random effects. The approach we propose below can be viewed as a generalization of an EIRM using only fixed effects. While there can be advantages to the use of random item parameters in some settings (De Boeck, 2008), the required distributional assumptions (normally distributed and symmetric about zero) may not be appropriate for item difficulties or for the theoretical processes believed to give rise to observed DIF (Halpin, 2024; Zhang et al., 2025). It is also our experience that the vast majority of DIF and MI analyses are performed in the fixed-items tradition (i.e., standard IRT and factor analysis models that place no distributional assumptions on item parameters), so this paper’s contribution can be viewed as bringing some of the affordances of the EIRM with respect to item covariates, to researchers using widely popular

fixed-items measurement models.

Penalized Approaches with Item Covariates. When dealing with item covariates, approaches based on penalized regression have also been used. Magis et al. (2015) proposed the use of LASSO regression for the detection of uniform DIF by applying the regularization penalty to the DIF term in the logistic regression method of Swaminathan and Rogers (1990). More recently, the LASSO penalties have been directly applied to DIF terms in item parameter decompositions in IRT and factor analytic models (e.g., Battauz, 2025; Belzak & Bauer, 2020; Wang et al., 2023). Outside of the DIF literature, regularization has also been used to aid maximum likelihood estimation in Rasch and IRT models (e.g., Battauz, 2020; Pan, 2018), including with explicit item covariate effects, namely item position effects (Kanopka & Domingue, 2025). It is also important to note that while Bayesian EIRM models with item covariates and EIRM models with random item effects can be considered types of regularization, making them conceptually similar to penalized regression approaches for DIF, these approaches are often performing parameter *shrinkage* as opposed to *selection* (with Bayesian models utilizing sparsity-inducing priors an important exception to this generalization).

In a DIF and MI context, the primary allure of regularized methods is that they can perform automated anchor item selection. That is, one does not need to select anchor items *a priori*; regularization using penalties that drive parameters to exactly zero can “pseudo-identify” a model so that anchor items emerge from DIF parameters being pushed to zero during model estimation. However, the performance of regularized DIF methods at smaller samples where measurement invariance analysis is often performed routinely is mixed, with relatively low power to detect DIF. This is somewhat by design, as the goal of regularization is to push parameters that contribute *relatively little* to model fit to zero, not just parameters that contribute *nothing at all* to model fit (Bauer et al., 2020; Belzak & Bauer, 2024; Lee et al., 2025). Regularized approaches thus serve somewhat different aims from the methods we describe in this paper, which are first and foremost about explaining

DIF in terms of what items have in common based on researchers’ understanding of item content, rather than trying to parse out the most important DIF items on purely empirical grounds with explanation of DIF done entirely *post hoc*.

Much of the work on regularized DIF analysis to date has been conducted in the general framework of MNLFA (e.g., Bauer et al., 2020; Belzak & Bauer, 2020, 2024; Brandt et al., 2025; S. M. Chen et al., 2022). While we do not focus on regularization in this paper, the approach we propose is based on MNLFA as well, as we believe it strikes a balance between the ability of EIRM-based approaches to incorporate item covariates into psychometric models and the potential pitfalls of unrealistic distributional assumptions that can confound interpretation of model results (Student, 2026). Additionally, the MNLFA-based approach is notably flexible with respect to person-side covariates, handling an arbitrary number of continuous or categorical (i.e., group-based) covariates (Kolbe et al., 2024; Molenaar, 2021), whereas prior work on regularized DIF analysis with item covariates is inherently group-based; this work also assumes a large item bank, something not available in most psychological research settings (Battauz, 2025), though as we demonstrate below, our approach does remain usable with large numbers of items. We present an overview of MNLFA next, before turning to our extension of this framework to handle DIF effects on groups of items.

Moderated Nonlinear Factor Analysis

MNLFA frames DIF and measurement noninvariance as types of *parameter moderation* (Bauer, 2017). Within the MNLFA framework, all of the parameters that make up a measurement model—latent variable mean(s) and factor (co)variance(s), as well as the item parameters—can potentially vary with an arbitrary number of background variables. In this way, EIRM utilizing only fixed effects can be seen as a restricted version of MNLFA (Bauer & Hussong, 2009). We begin by reviewing the generic formulation of a (unidimensional) MNLFA model, before turning to the decomposition of item effects that this study proposes.

Let $i \in \{1, \dots, N\}$ index persons, $j \in \{1, \dots, J\}$ index items, and $c \in \{1, \dots, C\}$ index covariates. Here y_{ij} encodes person i 's response to item j , and θ_i denotes person i 's level on the latent variable that causes item responses. We can write the distribution of the latent variable, conditional on covariates, as

$$\theta_i \sim \mathcal{N}(\alpha_i, \phi_i), \quad (1)$$

where the latent mean α_i and latent variance ϕ_i (the *impact* parameters) capture differences in the mean and variance of θ with respect to the C covariates (Bauer, 2017):

$$\alpha_i = \alpha_0 + \sum_{c=1}^C \alpha_c x_{ci} \quad (2)$$

$$\phi_i = \exp(\omega_0 + \sum_{c=1}^C \omega_c x_{ci}), \quad (3)$$

where x_{ci} is person i 's standing on covariate c . Equation 2 captures *mean impact*, the extent to which the mean of θ differs with each covariate. Equation 3 captures *variance impact* (i.e., heteroskedasticity; Kolbe et al., 2021, 2024), the extent to which the variance of θ differs with respect to the covariates after accounting for mean impact. We follow the general convention of modeling variance impact as log-linear to ensure that the variance is positive at all values of the covariates (Bauer, 2017).

The item response function for binary items modeled with a two-parameter logistic (2PL) IRT model in slope-intercept form can be written as

$$\Pr(y_{ij} = 1) = \text{logit}^{-1}(\lambda_{ji}\theta_i + \nu_{ji}), \quad (4)$$

where ν_{ji} is the item intercept and λ_{ji} is the item slope, again potentially varying across persons.¹

The item parameters for person i are written as functions of person i 's standing on

¹ Note that the slope-intercept parametrization is isomorphic with the difficulty-discrimination version of the 2PL, with the equivalencies $\lambda_{ji} = a_{ji}$ and $\nu_{ji} = -a_{ji}b_{ji}$.

each of the C covariates:

$$\nu_{ji} = \nu_{0j} + \sum_{c=1}^C \nu_{jc} x_{ci} \quad (5)$$

$$\lambda_{ji} = \lambda_{0j} + \sum_{c=1}^C \lambda_{jc} x_{ci}, \quad (6)$$

where ν_{jc} and λ_{jc} are the extent to which the item intercept and slope, respectively, change for item j and covariate c , relative to a baseline value ν_{0j} or λ_{0j} representing the value of the item parameter when all covariates take value 0. That is, ν_{jc} represents the extent to which item j becomes uniformly easier with each additional unit of covariate c , conditional on θ , while λ_{jc} represents the extent to which the item slope changes with each additional unit of c . This can be equivalently described as the extent to which the item becomes more or less discriminating with increasing values of the covariate; as a change in the responsiveness of the item to θ across values of the covariate; or as a three-way interaction between the item, latent variable, and covariate. ν_{jc} and λ_{jc} have also been described as capturing, respectively, *uniform* and *nonuniform* DIF with respect to covariate c (Student & McCormick, 2025). This approach easily generalizes to the functional forms specified by other measurement models from the IRT and factor analysis literatures. For instance, a linear factor analysis model can be modeled using Equations 1-3, 5, 6; all that changes is that instead of using the logit link function in Equation 4, one uses the identity link and models a residual. MNLFA for continuous indicators allows for covariate effects on this residual term as well (Bauer, 2017). Similarly, one can model covariate effects on the thresholds for ordered categorical items (Lee et al., 2024, 2025) in a graded response or (generalized) partial credit model (Masters, 1982; Muraki, 1992; Samejima, 1969), constrain the item slopes to 1 to produce a moderated Rasch model (Rasch, 1960), model covariate-moderated covariances between multiple latent variables (Bauer, 2017; Padgett, 2025), include a lower asymptote to create a moderated three-parameter logistic IRT model² (Bauer & Hussong, 2009; Birnbaum, 1968), and more. What defines an MNLFA

² Users should exercise extreme caution before attempting to decompose parameters with known

model, given this generality, is that any parameter of the measurement model can be written as a deterministic function of the covariates. Our concern in this paper is how the researcher specifies these functions for the item parameters; our focus in presentation is on the 2PL described above, but the approach can be employed for any measurement model that can be situated more broadly within the MNLFA framework and we provide an empirical example using a linear factor analysis model to emphasize this point.

Method: MNLFA with Item Group Effects

In the above parameterization of the moderated 2PL, each non-zero DIF parameter represents a unique item-specific (uniform or nonuniform) effect of the given covariate on the distribution of item responses. Here, we propose and study an extension to this model that parameterizes the effect of covariates on *groups of items*, with potential item-specific effects above and beyond these effects on sets of items. We view this as a route by which the researcher can bring theory about how the items measure the underlying construct across different levels of the covariates to bear on DIF/MI analysis. The goal is to support users as they test and build psychological theory, rather than automatically treating each item as having nothing in common with any other item (as in DIF analysis) or treating the entire testing enterprise as a hoop to jump through (as in MI analysis).

Model Specification

As above, let i index persons and j index items, but now also let $k \in \{1, \dots, K\}$ index groups of items and z_{kj} be a set of binary variables taking value 1 when item j is in group k , 0 otherwise. We can generalize equations 5 and 6 to decompose DIF into an effect at the item group level and an effect at the item-specific level, above and beyond the item

estimation issues such as the lower asymptote in 3PL IRT models. For more information on identifiability and DIF in the 3PL, see Maris and Bechger (2009) and Tay et al. (2011, 2013)

group, as:

$$\nu_{ji} = \nu_{0j} + \sum_{c=1}^C \left[\nu_{jc} x_{ci} + \sum_{k=1}^K \xi_{kc} z_{kj} x_{ci} \right] \quad (7)$$

$$\lambda_{ji} = \lambda_{0j} + \sum_{c=1}^C \left[\lambda_{jc} x_{ci} + \sum_{k=1}^K \kappa_{kc} z_{kj} x_{ci} \right]. \quad (8)$$

Equation 1 holds as above. Here, the item intercept and slope are now decomposed into:

- A baseline item parameter ν_{0j}, λ_{0j} , that is, the value of the parameter for the given item when all covariates take value zero.
- A set of item group-level covariate effects ξ_{kc}, κ_{kc} that apply identically to all items in the item group k , where item group membership is captured by binary variable $z_{kj} = 1$ when item j is in item group k ($z_{kj} = 0$ otherwise).
- An item-specific effect on each covariate above and beyond the item group-level effects ν_{jc}, λ_{jc} , applying only to this item for each covariate.

That is, we are first modeling DIF in the item parameters that is common to a *set* of items. Second, we (potentially) allow for DIF that is idiosyncratic to the given item and separate from any group-level effects.

Model Identification

MNLFA models, like all measurement models that aim to capture DIF, require anchor items for identification. These are items that are assumed DIF-free (with their DIF parameters fixed to zero), and that constitute the invariant scale on which θ is measured with respect to the covariate(s). The anchor items need not be the same across covariates or across parameters (i.e., an item can have DIF in the intercept but not the slope), but at least one anchor item per parameter-covariate combination is required if one also wants to estimate the impact parameters (that is, one cannot estimate DIF on every item and also estimate impact with respect to the same covariate; Bauer, 2017). Additionally, one must

set the scale for the latent variable, for example by fixing α_0 and ω_0 to 0 to yield $\theta_i \sim \mathcal{N}(0, 1)$ when all covariates take value 0.

These requirements become mildly more complex when item group effects are added into the model. The scale must be set as in the “standard” approach, but anchor item requirements are now imposed within item groups, not just across all of the items. This means that in order to identify the item group effect for a given parameter and item group, one must fix at least one item in that group’s item-specific DIF effect to zero; this is the same logic as the requirement of at least one anchor item to estimate impact is a standard MNLFA. One must also fix at least one of the item group effects for each parameter and covariate to identify the model; that is, one way to identify the scale is via an “anchor item group” in addition to the anchor item(s) *within* each group. In general, other approaches to model identification that are isomorphic with this approach, such as effect-coding the item groups instead of dummy-coding them, can be used instead for identification.

We also note that while a model with both item group effects on item parameters and item-specific effects within each item group *can* be identified conditional on anchor items, this does not mean that researchers *should* always estimate those item-specific effects. A model with all ν_{jc}, λ_{jc} fixed to zero, for instance, would be a model in which all DIF/noninvariance operates at the item group level. It is our expectation that much of the time, a theory-informed approach to DIF or MI analysis will start from this type of model. Additionally, the specification given above in equations 7, 8 is intended to be general but the choice of which parameter(s) to estimate is ultimately up to the researcher based on the theory they aim to test. Below we provide examples that do and do not fix the ν_{jc}, λ_{jc} to zero. When researchers want to choose among multiple theory-driven statistical models, comparisons of model-data fit are a foundational, though not foolproof, type of evidence for favoring one model over another (Bonifay & Cai, 2017). Although comparative fit indices broadly used in the structural equation modeling literature (e.g., Bentler, 1990; Muthén, 1993, etc.) are not available for MNLFA models (Bauer, 2017), MNLFA models estimated

via maximum likelihood can still be compared using information criteria (Burnham & Anderson, 2004), or via likelihood ratio tests when they are nested. Fortunately, it will often be the case that competing models are nested. For instance, if one models only item group effects, that is a more constrained version of a standard MNLFA where each DIF item gets its own DIF parameter estimate, allowing one to test models compatible with theories of one effect on items that form a group vs. an idiosyncratic effect on each item. We demonstrate this type of comparison below. Similarly, a standard MNLFA is nested within a model with both item group effects and item-specific DIF above and beyond item group effects. Additionally, though estimation is not the main focus of this paper, we note that this model can be straightforwardly estimated using marginal maximum likelihood (MML) and model constraints, or using Markov Chain Monte Carlo (MCMC), just as basic MNLFA models can be (Bauer, 2017; Enders et al., 2024).

Transparency and openness

The remainder of this paper consists of a motivating empirical example, an initial proof-of-concept simulation study, and three additional empirical examples intended to demonstrate the flexibility of the proposed approach. All data and analysis code are available via the Open Science Foundation (OSF), <https://osf.io/q9jbk>. These analyses were not preregistered. Data were analyzed using Mplus, version 9.0 (Muthén & Muthén, 2025), with file management and data cleaning conducted in R version 4.5.2 (R Core Team, 2025) using *MplusAutomation* (Hallquist & Wiley, 2018). The OSF repository also contains parallel examples of Bayesian implementations for the main simulation study and the Belief in Uearthly Beings analysis using Stan version 2.32.2 (Carpenter et al., 2017) and *rstan* version 2.32.7 (Stan Development Team, 2025).

Motivating Empirical Example: Gender Divergence in Development by Age

We illustrate the proposed approach by testing a specific theory regarding an instrument for measuring children’s development. We believe that the instrument may contain subsets of items that tap aspects of development that function nonequivalently

across boys and girls. Furthermore, these differences occur at a specific part of the intended age range such that scores for these children may systematically overstate or understate specific aspects of development by sex.

Data

The instrument we study is the Dutch Development Instrument (DDI), known previously as the Van Wiechen Schema (Jacobusse et al., 2006). This is a Dutch adaptation of the Bayley scales for infant development (see, e.g., Gagnon & Nagle, 2000) consisting of binary items describing specific behaviors and capabilities of the child. Data were originally collected by Herngreen et al. (1992) and coded such that a 1 means the child has demonstrated the described behavior while a 0 means the child has not. Jacobusse et al. (2006) evaluated and endorsed a unidimensional Rasch model for scaling the instrument, calling the resulting Rasch-based scale score the *D-score*. The DDI was revisited in the context of a larger global study developing the D-score as a purportedly interval-scaled and universal globally-comparable measure of the development of infants and toddlers (again based on the Rasch model; see McCray et al., 2023; Weber et al., 2019). The full 2019 study included data from a variety of developmental measurement instruments across low-, middle-, and high-income countries with the goal of establishing comparability across the assessments; though not described as such, it is an example of an integrative data analysis (Curran & Hussong, 2009), which is incidentally the setting in which MNLFAs were originally presented (Bauer & Hussong, 2009). The data we reanalyze were made public as part of this larger study; we accessed the data via the Item Response Warehouse (Domingue et al., 2025), where the dataset is named *dscore_dutch_weber_2019*, and they are also available from a repository linked in the 2019 study (Weber et al., 2019).

As outlined in Weber et al., 2019, DDI is intended to measure development from ages 0-2 years, but does not use the same items at each timepoint. Rather, the items used at each timepoint are tailored to the anticipated development of children at that age, so that the items are neither trivially easy nor exceedingly difficult, maximizing

developmental appropriateness and statistical information (i.e., reliability). At each timepoint, the instrument is made up of items from three developmental subdomains: gross motor, fine motor, and cognitive. The Rasch model outlined in the original study produces a scale score that is intended to be interpreted as a comparable measure of development irrespective of the child’s gender (or age, within the 0-24 months range).

In total, this study consisted of 9 waves of data collection, corresponding to child age windows ranging from 1 month to 24 months. For this example analysis we consider data from the eighth wave, corresponding to an age of 18 months³. A total of 1723 children have an observation in this wave. However, due to the original data collection design, there is a substantial amount of missing data (see Herngreen et al., 1992; Jacobusse et al., 2006; Weber et al., 2019) and inconsistency in which items each child was evaluated on. We focus on the 12 items for which the data are richest and a subsample of 706 children with full data on these items. The items are split evenly across the developmental domains of cognitive development, gross motor development, and fine motor development. For clarity of presentation we focus on this subsample, but we find near-identical results using full data and the same full-information marginal maximum likelihood approach used in the main analysis (see Appendix B).

Theory and Model Specification

The theory we aim to test in this analysis is that summarizing across both cognitive and fine/gross motor development items, effectively treating them as interchangeable indicators of development, is likely to systematically misrepresent development with respect to the sex of the child. Specifically, we test the hypothesis that relative to the cognitive items (the anchor item group), the fine and gross motor items (treated as separate item groups) will exhibit gender differences on their intercepts favoring boys.

The moderated Rasch model we consider, assuming conditional normality of θ for

³ We opted to use data from wave 8 instead of 9 because by wave 9, most of the items on the instrument are extremely easy, though analysis of earlier and later waves is certainly of interest.

marginal maximum likelihood estimation, is:

$$\theta_i \sim \mathcal{N}(\alpha_i, \phi) \quad (9)$$

$$\alpha_i = \alpha_0 + \alpha_1 x_{1i} \quad (10)$$

$$\Pr(y_{ij} = 1 \mid \theta_i) = \text{logit}^{-1}(\theta_i + \nu_{ji}) \quad (11)$$

$$\nu_{ji} = \nu_{0j} + \xi_{21} z_{2j} x_{1i} + \xi_{31} z_{3j} x_{1i} \quad (12)$$

Following the conventions for Equations 7 & 8, x_{1i} takes value 1 when the respondent is female and 0 otherwise, c indexes sex (where 1 indicates female) and $k = 2$ or $k = 3$ indexes item groups with cognitive items serving as the anchor group, $k = 2$ indicating that the item is a fine motor item, and $k = 3$ indicating that the item is a gross motor item. Thus, we are investigating the extent to which the two sets of motor items are harder or easier for girls than boys, above and beyond the differences in performance on the items for which DIF is not modeled (the cognitive items). If we fix $\xi_{21} = \xi_{31} = 0$, we get a “standard” (i.e., unmoderated item parameters) Rasch model with no item group effects. We estimate both of these models and compare them, both in terms of fit to the data and in terms of the substantive information the two models produce (or not) about sex differences in development at 18 months. To further demonstrate the value of estimating a common intercept DIF parameter across items within each group instead of just freely estimating DIF on an item-by-item basis, we also fit a third model in which DIF is separately estimated for each item in the motor item groups, i.e., a standard model for DIF in the intercepts of all fine and gross motor items.

Recognizing that Rasch analysis is sometimes viewed with general skepticism among practitioners more inclined toward IRT or factor analysis (Andrich, 2004), we replicate this analysis in the context of the slope-intercept 2PL model outlined above. Here, we again estimate item group effects on the intercepts of the fine and gross motor items relative to an anchor item group consisting of cognitive items. The difference is that we now allow the item *slopes* to vary instead of fixing them all to 1 as in a Rasch parameterization. To

ensure comparability of the results across all four models, we present unstandardized results as well as standardized results that fix the full variance of θ to 1 in each model, rendering the model coefficients directly comparable despite the different parameterizations and approaches to setting the scale in the original models. The moderated 2PL from which we start is:

$$\theta_i \sim \mathcal{N}(\alpha_i, 1) \quad (13)$$

$$\alpha_i = \alpha_0 + \alpha_1 x_{1i} \quad (14)$$

$$\Pr(y_{ij} = 1 \mid \theta_i) = \text{logit}^{-1}(\lambda_{ji} \theta_i + \nu_{ji}) \quad (15)$$

$$\nu_{ji} = \nu_{0j} + \xi_{21} z_{2j} x_{1i} + \xi_{31} z_{3j} x_{1i} \quad (16)$$

$$\lambda_{ji} = \lambda_{0j} + \kappa_{21} z_{2j} x_{1i} + \kappa_{31} z_{3j} x_{1i} \quad (17)$$

For comparison, we fit two additional models in this case: one in which $\kappa_{21} = \kappa_{31} = 0$, imposing full invariance of the item slopes but allowing for item group DIF on the intercepts, and one in which we fix $\kappa_{21} = \kappa_{31} = \xi_{21} = \xi_{31} = 0$, producing a “standard” slope-intercept 2PL with a latent regression on sex. Note that we do not present models that include terms for heteroskedasticity; this is because initial models including these terms did not yield any additional information or qualitatively change any conclusions, and the models with a constant residual θ variance are more straightforward to present.

We estimate each model via full-information marginal maximum likelihood as implemented in Mplus⁴. All model scripts, as well as code to clean the data in R and cleaned Mplus data files, are available via OSF.

⁴ We use the standard “ML” estimator for this example as this estimator allows for straightforward likelihood ratio testing that we believe is valuable for illustration purposes because it is replicable from the loglikelihood and parameter counts alone; results using MLR and scaling corrections (Satorra & Bentler, 2010) were qualitatively identical, with the LRT still rejecting the unmoderated model for the moderated model but not rejecting the moderated model for the full DIF model.

Model Results

Rasch parameterization

We find evidence that the model with moderation of motor items' intercepts by sex fits the data significantly better than a latent regression Rasch model. Table 1 presents this comparison in terms of information criteria and a likelihood ratio test among the nested models; all evidence points to the model with moderation of the item intercepts by sex fitting better than the latent regression model. Additionally, the model with item group-level intercept moderation is favored relative to a model with item-specific DIF parameters for each motor item in terms of AIC and BIC, and the likelihood ratio test does not provide evidence that this model fits better than the model with item group-level moderation.

Table 1

Fit comparison of different moderated Rasch models for child development.

Model	Loglikelihood	Parameters	AIC	BIC	LRT p-value
Latent Regression	-3376.35	14	6781	6845	–
Item Group Moderated	-3361.69	16	6755	6828	0.000
Full DIF	-3357.59	22	6759	6859	0.224

Note. AIC = Akaike Information Criterion, BIC = Bayesian Information Criterion, LRT p-value = p-value based on likelihood ratio test of difference in model fit via χ^2 distribution with degrees of freedom equal to difference in number of parameters for adjacent models. **Latent Regression** = standard Rasch model with regression on sex; **Item Group Moderated** = Rasch model with one parameter for intercept DIF across all fine motor items and one parameter for intercept DIF across all gross motor items; **Full DIF** = Rasch model with DIF separately estimated for each gross and fine motor item.

We report the model parameters that shed substantive light on development with

respect to child sex at 18 months in Table 2; here we only compare the latent regression model to the model with item group-level intercept DIF as the model with item-specific DIF did not fit better than the model with item group level DIF. Parameter α_1 in the latent regression model represents the mean difference in development between boys and girls in the sample, assuming that the items are interchangeable and invariant measures of development. We can see here that we cannot reject the null hypothesis of no mean difference by sex. However, the item group-moderated model tells a substantively different, and on our view informative, story. In this model α_1 now represents means impact treating only the cognitive items as invariant across sex. Parameters ξ_{21} , ξ_{31} now capture a constant effect in logits on the intercept of the fine and gross motor items, respectively. We can see that girls appear to significantly outpace boys in development based on the cognitive items, but that when we set the scale this way, there is significant DIF on both types of motor item favoring boys. Substantively this tells us that in this population, for a given level of development as estimated by the primary Rasch model, we are likely to be overstating boys' cognitive development and understating their motor development, while the opposite holds for girls.

Table 2

Relevant parameter estimates for moderated Rasch models for child development.

Model	Parameter	Interpretation	Estimate (SE)	Standardized (SE)
Latent Regression	α_1	Difference in mean development for female vs. male children	0.220 (0.118)	0.171 (0.091)
Item Group Moderated	α_1	Difference in mean development for female vs. male children <i>as</i> <i>represented by cognitive</i> <i>items (anchor item</i> <i>group)</i>	0.625 (0.145)	0.471 (0.104)
	ξ_{21}	Intercept effect on fine motor items	-0.501 (0.160)	-0.501 (0.160)
	ξ_{31}	Intercept effect on gross motor items	-0.846 (0.159)	-0.846 (0.159)

2PL version

A 2PL analysis leads to the same substantive conclusions. Here we present models with no item parameter moderation, moderation of intercepts at the item group level, and moderation of both intercepts and slopes at the item group level. The model with just moderation of the intercepts is favored by all evaluation metrics (although we note that BIC actually favors the Rasch parameterization). See Table 3 for fit information and Table A1 for relevant parameter estimates; data and scripts for full replication are available in the OSF. At a high level, we again find evidence that a fully invariant solution does not capture substantial differences in cognitive vs. motor development between boys and girls.

Table 3*Fit comparison of moderated vs. unmoderated 2PL models for child development.*

Model	Loglikelihood	Parameters	AIC	BIC	LRT p-value
Latent Regression	-3354.83	25	6760	6874	–
Moderated Intercepts	-3344.03	27	6742	6865	0.000
Moderated Intercepts and Slopes	-3341.76	29	6742	6874	0.103

Note. AIC = Akaike Information Criterion, BIC = Bayesian Information Criterion, LRT p-value = p-value based on likelihood ratio test of difference in model fit via χ^2 distribution with two degrees of freedom, comparing adjacent models (i.e., moderated intercepts vs. latent regression, moderated intercepts and slopes vs. moderated intercepts only).

Model Interpretation

Importantly, our approach does not simply lead to the conclusion that measurement invariance is violated and scores cannot be compared across boys and girls (Rohrer & Paulewicz, 2025), nor does it automatically suggest that the gender differences across item subconstructs are indications of gender *bias* in the sense of the items capturing construct-irrelevant gender-specific variance (Belzak, 2023; Camilli, 1993). Rather, what we now know is that aggregating item responses at 18 months into a single composite score (via the Rasch or 2PL model) absorbs gender differences on cognitive vs. motor skills in a way that essentially cancels out, giving the impression that overall development is comparable among boys and girls in this population when in fact the specific areas where children develop appear to diverge meaningfully by sex. Importantly, this does not render the originally scores meaningless (Robitzsch & Lüdtke, 2023); rather, it illustrates information lost by the use of a scoring model that is specified as invariant across sex. Whether or not this loss of information is of concern is a matter of the use of the score (AERA et al., 2014). For example, aggregated comparisons of development across geographic regions would be unlikely to be misleading if development is treated as invariant

because the proportion of boys vs. girls would not generally be expected to vary much from one setting to the next. On the other hand, use of the D-score for studying sex differences in development during infancy without accounting for differences in motor vs. cognitive development might systematically lead to the conclusion that boys are more developed in terms of cognitive milestones, and less developed in terms of motor milestones, than they really are as they approach early childhood. This analysis equips the researcher with the knowledge needed to make a principled decision about absorbing these subconstruct-level sex differences into a single overall composite.

Simulation Study: Identification of DIF Effects on Subsets of Items

Given the results above, we next turn to a simulation study to generate evidence on the usability of this modeling approach in practice. Specifically, we want to ensure that (a) if effects on item groups are present and the model is correctly specified, they can be recovered with predictable statistical properties (i.e., power and precision that increase with person sample size, number of items in each group, magnitude of the effects; consistent confidence interval coverage; minimal bias), whether or not there are also impact effects on the latent variables, as well as that (b) if effects on item groups are not present, the model will not erroneously produce evidence of such effects. We view this as foundational but non-exhaustive evidence of the practical usability of the methods outlined above. To keep this first study manageable, we do not consider item-specific effects above and beyond item group effects; we consider this in an empirical example below. We also limit our study to DIF on item intercepts, leaving simulation work on the recovery of item slope DIF effects as a future study, given that intercept DIF most directly affects comparison of scores among individuals and groups who differ on covariates or aggregated comparisons across those covariates (Halpin, 2024; Halpin & Gilbert, 2024).

Study Design

We fully cross the following variables:

- Person sample size: 250, 500, 1000, 2000

These sample sizes cover a wide range of common sample sizes in psychological research.

- Number and type of covariates: one binary covariate (e.g., group membership), or two correlated covariates, one binary and one continuous

Many covariates, especially in traditional DIF applications, represent a contrast between two groups with respect to a single demographic variable (e.g., race or gender), so we study performance of the proposed method in this scenario. To keep the simulation conditions manageable but to also capture one of MNLFA's main differentiators compared to traditional and modern group-based methods, we also consider a scenario with both a binary and a continuous background variable. In all conditions, the binary background variable is split evenly in the sample. In conditions with a second (continuous) background variable, the continuous variable is generated according to the following normal distribution, where x_{1i} is person i 's value on the binary covariate (0 or 1) and x_{2i} is their value on the continuous covariate:

$$x_{2i} \sim \mathcal{N}(x_{1i} - 0.5, 1), \quad (18)$$

yielding a continuous covariate with $\mathbb{E}[x_{1i}] = 0$, $\sigma_{x_2}^2 = 1.25$, covariance with x_{1i} of $\sigma_{x_1x_2} = 0.25$, and correlation $r(x_{1i}, x_{2i}) \approx 0.4472$. We use this data-generating condition because the primary reason to consider multiple covariates in the same model in a DIF context is that correlated covariates can lead to misattribution of DIF to the wrong covariate if they are modeled separately. If the covariates are uncorrelated, it only becomes easier for the model to recover DIF with respect to the correct covariate, so this more challenging condition is more directly informative for practice.

- Number of items per item group: 4, 10

This helps us understand the extent to which power and statistical precision are improved with larger item groups, while also recognizing that psychological and educational instruments vary greatly in length.

- Number of item groups: 3, 6

Similarly, we vary the number of item groups to understand the extent to which more or fewer item groups lead to better or worse parameter recovery, and to produce instrument lengths that vary from 12 (in line with a short psychological survey instrument) to 60 (in line with a large-scale educational test).

- Magnitude of item-specific effects: small or large

We simulate DIF effects at the item group level such that the first item group has zero DIF on any item with respect to the first binary covariate, the second has intercept DIF that is positive, and the third has intercept DIF that is negative. If there are six item groups, this pattern is repeated for item groups 4-6. For the small DIF condition, the magnitude of the effect is $\xi_{21} = 0.1$ for item group 2 and $\xi_{31} = -0.1$ for item group 3. For the large DIF condition, the magnitudes are $\xi_{21} = 0.5$ and $\xi_{31} = -0.5$.

In the case of two covariates, the effects with respect to the binary covariate stay the same, and we add in linear effects of the continuous covariate on item intercepts as well in the same direction. For the small DIF case, these effects take value $\xi_{22} = 0.05$ and $\xi_{32} = -0.05$. For the large DIF case, they take value $\xi_{22} = 0.2$ and $\xi_{32} = -0.2$.

- Magnitude of impact on latent variable: small or large

We do not study zero-impact conditions because covariate impact of zero is rare in practice (Bauer, 2017, 2023) and because impact alongside DIF is a more challenging condition for empirical identification than DIF without impact. We define impact as small according to the equation,

$$\theta_i \sim \mathcal{N}(0.2x_{1i} + 0.1x_{2i}, \exp(0.1x_{1i} + 0.05x_{2i})), \quad (19)$$

and large according to the equation,

$$\theta_i \sim \mathcal{N}(0.5x_{1i} + 0.25x_{2i}, \exp(0.4x_{1i} + 0.2x_{2i})), \quad (20)$$

where x_{2i} takes value 0 for all persons in the one-covariate condition, effectively removing it from the model. Thus θ is standard normal when both covariates are 0.

Items are all binary, and are generated using fixed baseline parameters (i.e., the item parameters when covariates have value 0). Within each item group, we space item intercepts evenly, in order, from -1.5 to 1.5. Within the first and second halves of each item group, we space item slopes evenly from 0.8 to 1.2. Thus in a group of 10 items, the intercepts would be -1.5, -0.5, 0.5, 1.5, and the corresponding slopes would be 0.8, 1.2, 0.8, 1.2. With ten items per group the intercepts would be -1.500, -1.167, -0.833, -0.500, -0.167, 0.167, 0.500, 0.833, 1.167, 1.500. The slopes would be 0.8, 0.9, 1.0, 1.1, 1.2, 0.8, 0.9, 1.0, 1.1, 1.2.

We repeat this 250 times for each combination of design variables, of which there are 128, producing a total of 32,000 datasets. Next we describe the model we fit to each dataset.

Primary Model Fitting and Evaluation

We estimate the following model with one covariate:

$$\theta_i \sim \mathcal{N}(\alpha_i, \phi_i) \quad (21)$$

$$\alpha_i = \alpha_1 x_{1i} \quad (22)$$

$$\phi_i = \exp(\omega_1 x_{1i}). \quad (23)$$

That is, the scale of the model is set by fixing θ to standard normal when the covariate takes value 0. The item response function is as follows where C is the number of covariates (1 or 2), J is the number of items (ranging from 12 to 60 depending on item group size and count), and K is the number of item groups (3 or 6):

$$\Pr(y_{ij} = 1 \mid \theta_i) = \text{logit}^{-1}(\lambda_j \theta_i + \nu_{ji}) \quad (24)$$

$$\nu_{ji} = \nu_{0j} + \sum_{c=1}^C \sum_{k=2}^K \xi_{kc} z_{kj} x_{ci}. \quad (25)$$

As noted we do limit this study to intercept DIF, so slopes are treated as invariant across covariates. Note as well that we treat the first item group as the anchor item group,

and that we estimate DIF parameters for item group 4 in the six item group condition even though the true value is 0. This allows us to understand both power to detect effects that exist and susceptibility to Type I error in the same model.

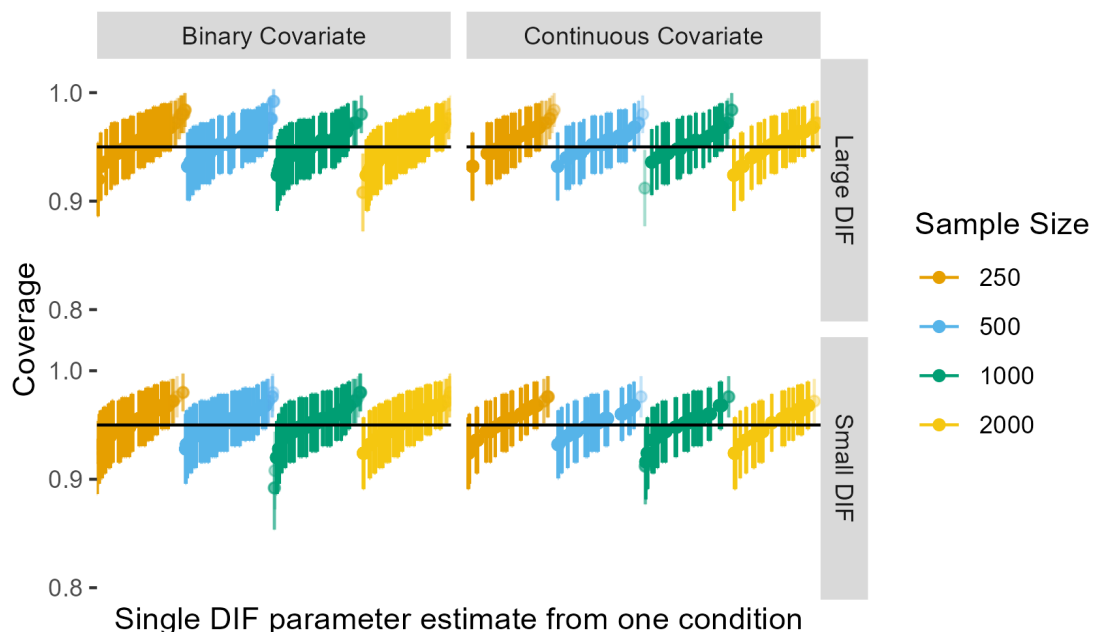
We report traditional parameter recovery evaluation statistics: bias, 95% confidence interval coverage, null hypothesis rejection rate (interpreted as power when an effect is present and Type I error rate when it is not), and standard error bias (mean standard error minus standard deviation of estimates). For bias and coverage, we compute Monte Carlo standard errors and use these standard errors to plot confidence intervals for all relevant parameter estimates' bias and coverage.

Results

DIF Effect Recovery

At a high level, recovery of the item group-level DIF effects appears to follow predictable patterns: larger samples and larger effects lead to more certainty in parameter estimates, with nominal coverage rates and minimal standard error bias across conditions. This suggests no systematic issues with the empirical identification of the modeled effects when the model is correctly specified. Figures 1 and 2 show this result graphically.

Table 4 summarizes the simulation results more broadly, disaggregated for DIF effects that are zero vs. non-zero. For nonzero parameters, there is a predictable increase in power as both effects and sample size grow, with power of .78 or above for large DIF effects at $N = 500$ but as low as 0.09 in one case where DIF is small even with $N = 1000$. Importantly, rejection rates when true DIF is zero (i.e., type I error rates) are well-controlled in all cases.

**Figure 1**

95% Confidence Interval Coverage with Monte Carlo Standard Error for All DIF Parameters across All Conditions. Note. Each point summarizes a single parameter estimate from one condition. A transparent confidence interval indicates the the Monte Carlo confidence interval does not include 0.95.

Table 4

Summary Simulation Results for Item Group DIF Parameters

Eff.	=	Sample	Covariate Magnitude	Bias	Coverage	SE Bias	Rej. Rate	
0?		Size						
No		250	Bin.	Small	-0.04-0.03	0.92-0.98	-0.03-0.03	0.04-0.14
No		250	Bin.	Large	-0.04-0.04	0.92-0.98	-0.02-0.02	0.46-0.96
No		250	Cont.	Small	-0.01-0.01	0.92-0.98	-0.00-0.01	0.04-0.14
No		250	Cont.	Large	-0.02-0.02	0.94-0.98	-0.00-0.01	0.36-0.85
No		500	Bin.	Small	-0.01-0.02	0.93-0.98	-0.01-0.01	0.06-0.25

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Table 4 – Continued from previous page

Eff. 0?	= Sample Size	Covariate	Magnitude	Bias	Coverage	SE Bias	Rej. Rate
No	500	Bin.	Large	-0.02-0.02	0.93-0.99	-0.01-0.02	0.78-1.00
No	500	Cont.	Small	-0.01-0.01	0.93-0.98	-0.00-0.00	0.07-0.24
No	500	Cont.	Large	-0.01-0.01	0.93-0.98	-0.00-0.01	0.65-0.99
No	1000	Bin.	Small	-0.02-0.02	0.89-0.98	-0.02-0.01	0.09-0.40
No	1000	Bin.	Large	-0.02-0.02	0.92-0.98	-0.01-0.01	0.98-1.00
No	1000	Cont.	Small	-0.00-0.00	0.91-0.98	-0.00-0.00	0.13-0.38
No	1000	Cont.	Large	-0.00-0.01	0.91-0.98	-0.00-0.01	0.92-1.00
No	2000	Bin.	Small	-0.01-0.01	0.92-0.98	-0.00-0.01	0.18-0.60
No	2000	Bin.	Large	-0.01-0.01	0.91-0.98	-0.01-0.01	1.00-1.00
No	2000	Cont.	Small	-0.00-0.00	0.92-0.97	-0.00-0.00	0.22-0.63
No	2000	Cont.	Large	-0.00-0.00	0.92-0.97	-0.00-0.00	0.99-1.00
Yes	250	Bin.	Small	-0.01-0.02	0.93-0.97	-0.01-0.03	0.03-0.07
Yes	250	Bin.	Large	-0.04-0.02	0.94-0.98	0.00-0.02	0.02-0.06
Yes	250	Cont.	Small	-0.01-0.01	0.94-0.98	-0.00-0.01	0.02-0.06
Yes	250	Cont.	Large	-0.01-0.00	0.93-0.98	-0.00-0.01	0.02-0.07
Yes	500	Bin.	Small	-0.01-0.02	0.94-0.98	-0.01-0.01	0.02-0.06
Yes	500	Bin.	Large	-0.02-0.01	0.93-0.97	-0.00-0.01	0.03-0.07
Yes	500	Cont.	Small	-0.00-0.00	0.94-0.96	-0.00-0.00	0.04-0.07
Yes	500	Cont.	Large	-0.00-0.01	0.94-0.96	-0.00-0.00	0.04-0.06
Yes	1000	Bin.	Small	-0.01-0.01	0.94-0.98	-0.00-0.01	0.02-0.06
Yes	1000	Bin.	Large	-0.02-0.01	0.94-0.96	-0.01-0.00	0.04-0.06

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Table 4 – Continued from previous page

Eff. = 0?	Sample Size	Covariate	Magnitude	Bias	Coverage	SE Bias	Rej. Rate
Yes	1000	Cont.	Small	-0.00-0.00	0.94-0.97	-0.00-0.00	0.03-0.06
Yes	1000	Cont.	Large	-0.00-0.00	0.94-0.97	0.00-0.01	0.03-0.06
Yes	2000	Bin.	Small	-0.00-0.01	0.94-0.97	-0.01-0.00	0.03-0.06
Yes	2000	Bin.	Large	-0.01-0.01	0.92-0.98	-0.00-0.01	0.02-0.08
Yes	2000	Cont.	Small	0.00-0.00	0.92-0.94	-0.00-0.00	0.05-0.08
Yes	2000	Cont.	Large	-0.00-0.00	0.94-0.97	-0.00-0.00	0.03-0.06

Note. **Eff. = 0?** = whether data-generating DIF effect was zero or nonzero. Rejection rate for nonzero effects is power; for zero effects it is type I error rate. **Magnitude** refers to magnitude of DIF effects.

Mean and Variance Impact Parameter Recovery

Mean and variance impact recovery exhibit no systematic bias and nominal coverage. Table 5 presents summary statistics for mean impact by sample size and true impact magnitude, while Table 6 presents these results for variance impact. Bias and standard error bias are both minimal, which in turns tells us that the root-mean-square error of the estimates is in all cases very close to the sampling distribution of the relevant parameter. When mean impact is large, power to detect it is fairly high, exceeding 0.8 in some conditions with sample sizes as small as 250, while larger samples are needed to detect smaller mean impact effects. Variance impact requires larger samples for consistent detection. Figures 3 and 4 show that 95% confidence interval coverage is generally very close to nominal, while Figure 5 shows that bias is generally minimal with almost all Monte Carlo confidence intervals for bias containing zero.

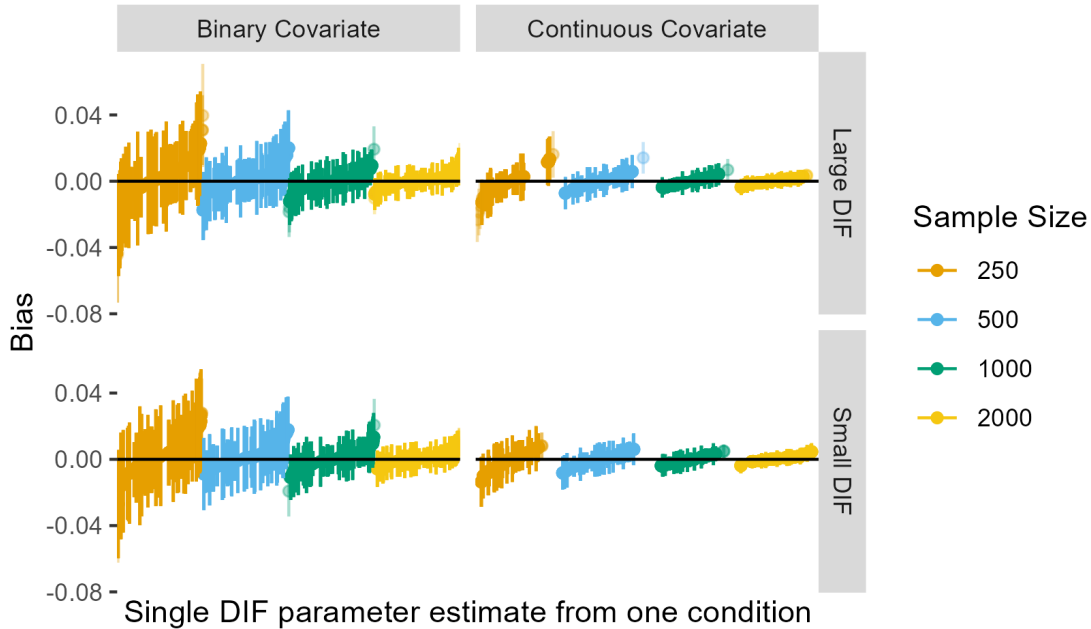


Figure 2

Bias with Monte Carlo Standard Error for All DIF Parameters across All Conditions.

Note. Each point summarizes a single parameter estimate from one condition. A transparent confidence interval indicates the the Monte Carlo confidence interval does not include 0.

Baseline Item Parameter Recovery

While not the primary focus of the study, recovery of baseline item parameters is also excellent, with nominal coverage, precision that improves with sample size, and minimal standard error bias (which in turns indicates bias close to zero and mean-square error close to sampling variance). Results for item parameter recovery are included in Appendix C, Table C1.

Summary

The above results tell us that empirical identification of item group-level DIF effects on item intercepts is straightforward with maximum likelihood estimation, and that standard errors are valid. Power is predictable, with larger samples and larger effects leading to greater power, while type I error is well-controlled.

Table 5*Summary Simulation Results for Mean Impact*

SS	Covariate	Impact	Bias	Coverage	SE Bias	Rej. Rate
250	Bin.	Large	-0.03-0.03	0.94-0.99	-0.00-0.03	0.49-0.83
250	Bin.	Small	-0.03-0.01	0.94-0.99	-0.00-0.04	0.06-0.20
250	Cont.	Large	0.01-0.02	0.95-0.97	-0.00-0.01	0.60-0.83
250	Cont.	Small	-0.00-0.02	0.96-0.99	-0.00-0.02	0.12-0.21
500	Bin.	Large	-0.02-0.02	0.94-0.98	-0.01-0.01	0.82-0.98
500	Bin.	Small	-0.02-0.01	0.94-0.98	-0.00-0.02	0.18-0.39
500	Cont.	Large	-0.00-0.01	0.93-0.97	-0.00-0.01	0.92-1.00
500	Cont.	Small	-0.00-0.01	0.95-0.97	-0.00-0.01	0.24-0.41
1000	Bin.	Large	-0.01-0.01	0.93-0.97	-0.01-0.01	0.98-1.00
1000	Bin.	Small	-0.01-0.01	0.92-0.97	-0.00-0.01	0.40-0.70
1000	Cont.	Large	-0.00-0.01	0.93-0.97	-0.00-0.00	1.00-1.00
1000	Cont.	Small	-0.00-0.01	0.94-0.98	-0.00-0.01	0.49-0.77
2000	Bin.	Large	-0.01-0.01	0.92-0.97	-0.00-0.00	1.00-1.00
2000	Bin.	Small	-0.01-0.01	0.92-0.99	-0.00-0.00	0.70-0.94
2000	Cont.	Large	-0.00-0.00	0.92-0.98	-0.00-0.00	1.00-1.00
2000	Cont.	Small	-0.00-0.00	0.93-0.98	-0.00-0.00	0.81-0.96

Note. **Covariate:** Bin. = binary, Cont. = continuous.

Additional Empirical Examples

Equipped with evidence that our modeling approach can recover the type of effects it is intended to (but has well-controlled Type I error rates when there are no effects present), we present additional empirical examples to further illustrate the flexibility and applicability of our approach. These examples also provide the chance to demonstrate

Table 6*Summary Simulation Results for Variance Impact*

SS	Covariate	Impact	Bias	Coverage	SE Bias	Rej. Rate
250	Bin.	Large	-0.05-0.03	0.93-0.97	-0.02-0.04	0.20-0.56
250	Bin.	Small	-0.04-0.02	0.92-0.98	-0.02-0.02	0.04-0.12
250	Cont.	Large	-0.01-0.02	0.93-0.96	0.00-0.02	0.23-0.42
250	Cont.	Small	-0.01-0.01	0.95-0.98	0.01-0.03	0.04-0.07
500	Bin.	Large	-0.01-0.02	0.92-0.98	-0.01-0.01	0.38-0.81
500	Bin.	Small	-0.03-0.01	0.93-0.98	-0.01-0.03	0.04-0.12
500	Cont.	Large	-0.00-0.01	0.93-0.96	-0.00-0.01	0.43-0.79
500	Cont.	Small	-0.01-0.02	0.95-0.98	0.00-0.01	0.04-0.11
1000	Bin.	Large	-0.01-0.02	0.90-0.98	-0.01-0.01	0.67-0.98
1000	Bin.	Small	-0.02-0.01	0.92-0.97	-0.01-0.01	0.08-0.18
1000	Cont.	Large	-0.00-0.01	0.92-0.96	-0.00-0.00	0.76-0.98
1000	Cont.	Small	-0.01-0.01	0.94-0.97	-0.00-0.01	0.07-0.15
2000	Bin.	Large	-0.01-0.01	0.92-0.98	-0.01-0.01	0.94-1.00
2000	Bin.	Small	-0.01-0.01	0.92-0.97	-0.01-0.01	0.13-0.28
2000	Cont.	Large	-0.00-0.01	0.92-0.97	-0.00-0.00	0.97-1.00
2000	Cont.	Small	-0.00-0.00	0.93-0.97	-0.00-0.00	0.16-0.31

Note. **Covariate:** Bin. = binary, Cont. = continuous.

several of the affordances of the proposed approach in the context of studies covering a variety of psychological constructs. The primary focus of these examples is modeling flexibility, rather than on generating strong theories to substantiate the models whose estimation we demonstrate.



Single mean impact estimate from one condition

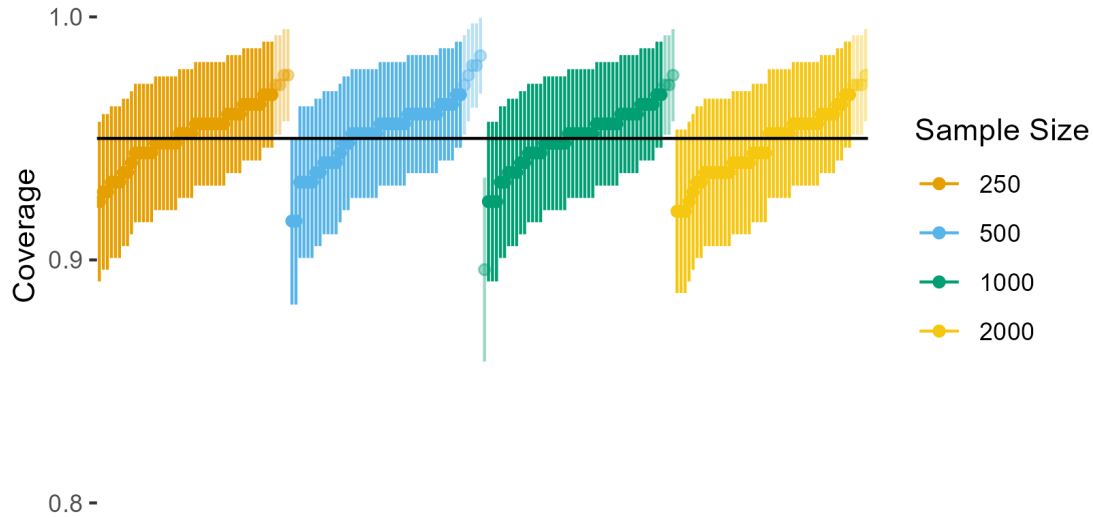
Figure 3

95% Confidence Interval Coverage with Monte Carlo Standard Error for All Mean Impact Parameters across All Conditions. Note. Each point summarizes a single parameter estimate from one condition. A transparent confidence interval indicates the Monte Carlo confidence interval does not include 0.95.

Item-Specific DIF Above and Beyond Item Group Effects

Overview

Bohn et al., 2025 present an instrument for measuring vocabulary development in 3-8 year olds in Germany. The Parent Report of Expressive Vocabulary in Children (PREVIC) consists of a set of words, each of which constitutes an item for which a response of 1 indicates that the child being considered has ever used the word, while a 0 indicates that the child has never used the word. The item pool is made up of adjectives, nouns, and verbs that vary substantially in difficulty, and the paper reports a set of analyses focused on reducing a large set of 379 candidate items to a final set of 89 based on responses from 1190 respondents. The authors report no evidence of gender-based DIF in their final set of items. We revisit their openly available data (named *previc_bohn_2023* in



Single variance impact estimate from one condition

Figure 4

95% Confidence Interval Coverage with Monte Carlo Standard Error for All Variance Impact Parameters across All Conditions. Note. Each point summarizes a single parameter estimate from one condition. A transparent confidence interval indicates the the Monte Carlo confidence interval does not include 0.95.

the Item Response Warehouse) to consider DIF at the word *type* level instead of at the individual item level.

We use response data from all 1190 respondents; respondents here are parents of children who range in age from 3-8. Among the 379 items in the full pool of potential PREVIC items, we retain all items with endorsement rates below 95%, as these exceedingly easy items contain essentially no statistical information. This leaves 267 items, of which 62 are adjectives, 68 are verbs, and 137 are nouns. The model we specify here demonstrate the simultaneous modeling of item group and item-specific effects by considering DIF with respect to the dataset's gender variable, coded 0 = male, 1 = female.

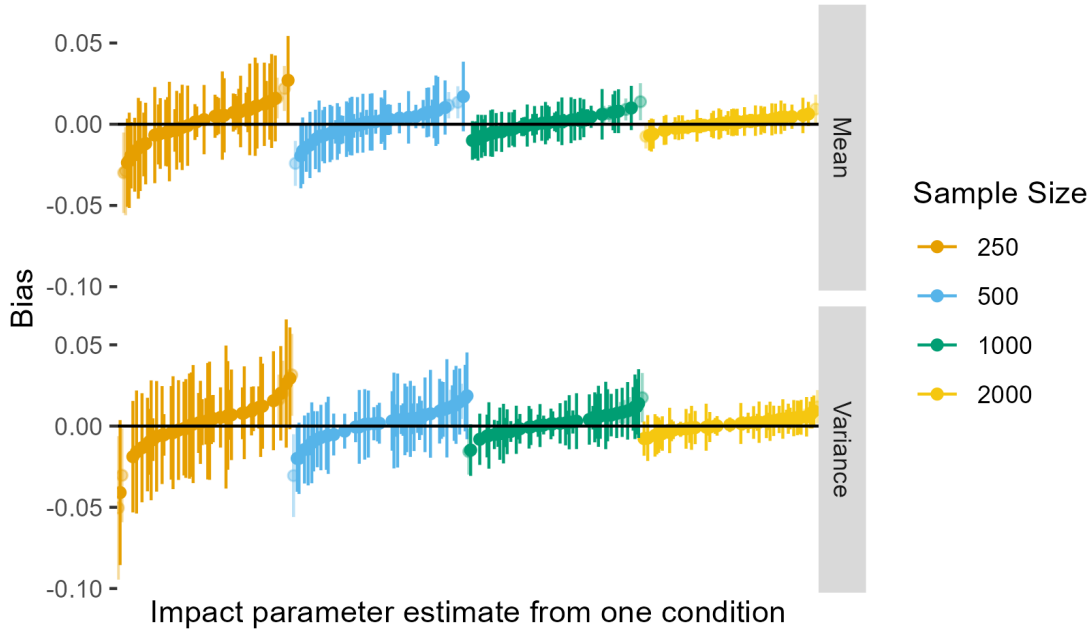


Figure 5

Bias with Monte Carlo Standard Error for All Impact Parameters across All Conditions.

Note. Each point summarizes a single parameter estimate from one condition. A transparent confidence interval indicates the the Monte Carlo confidence interval does not include 0.

Model Specification

We use the 2PL, as in the other examples with binary item responses. We specify three item groups: nouns, adjectives, and verbs. The anchor item group is nouns, and we estimate item group-level intercept DIF effects for verbs and adjectives. Previewing our findings, we find that there is no gender effect for verbs, but that girls appear to significantly outperform boys on adjectives conditional on their noun performance (i.e., girls in this population acquire adjectives faster than boys do relative to their overall vocabulary). In the context of this model, we test the hypotheses that two specific adjectives exhibit a further gender DIF effect *above and beyond* this main effect on adjectives as a whole. The words are *mager* and *treu*, which translate respectively to *skinny* and *faithful* (or *true*).

We also fit a latent regression model assuming full invariance against which to test the fit of the models with DIF terms. Mplus scripts for all models are available via the OSF. Noting that *mager* is item 7 in our cleaned dataset (available in the OSF as well) and *treu* is item 54, the full model specification is:

$$\theta_i \sim \mathcal{N}(\alpha_i, \phi_i), \quad (26)$$

where the latent mean α_i and latent variance ϕ_i (the *impact* parameters) capture differences in the mean and variance of θ with respect to the C covariates (Bauer, 2017):

$$\alpha_i = \alpha_1 x_{1i} \quad (27)$$

$$\phi_i = \exp(\omega_1 x_{1i}) \quad (28)$$

$$\Pr(y_{ij} = 1 \mid \theta_i) = \text{logit}^{-1}(\lambda_j \theta_i + \nu_{ji}) \quad (29)$$

$$\nu_{ji} = \nu_{0j} + \nu_{1j} x_{1i} + \xi_{21} z_{2j} x_{1i} + \xi_{31} z_{3j} x_{1i}. \quad (30)$$

Here, x_{1i} , the only covariate, refers to the binary-coded gender variable for person i , while z_{2j} indicates that item j is an adjective while z_{3j} indicates a verb. Note that we are treating all item slopes as invariant in this example; that we are estimating both mean and variance impact; and that we set the scale by fixing the θ distribution to standard normal for boys. We fix $\nu_{1j} = 0$ for all $j \notin \{7, 54\}$, while those two terms are freely estimated and capture item-specific DIF above and beyond the main adjective effect.

Results

We find empirical support for a gender effect favoring girls on adjectives as a whole as well as the two words for which we estimated item-specific additional effects. We find no evidence of an analogous effect for verbs. Key parameter estimates are found in Table 7, while fit information favoring this model over a latent regression with full invariance are found in Table 8. Our emphasis here is just on simultaneous modeling of item group and item-specific effects, but removing the non-significant verb effect would likely improve

model fit further ⁵.

Table 7

Key parameter estimates from a model of gender DIF in German vocabulary acquisition.

Parameter Type	Parameter	Estimate (SE)	Brief Interpretation
Mean Impact	α_1	0.031 (0.057)	Girls’ average vocabulary acquisition relative to boys’
Log-Variance Impact	ω_1	-0.121 (0.085)	Girls’ variability in vocabulary acquisition relative to boys’
Item Group DIF	ξ_{21}	0.177 (0.028)	Intercept DIF for girls relative to boys on adjectives
	ξ_{31}	0.035 (0.026)	Intercept DIF for girls relative to boys on verbs
Item Specific DIF	$\nu_{7,1}$	0.153 (0.034)	Intercept DIF for girls relative to boys on <i>mager</i> residual of adjective effect
	$\nu_{54,1}$	0.404 (0.029)	Intercept DIF for girls relative to boys on <i>treu</i> residual of adjective effect

Continuous Factor Indicators: Belief in Unearthly Beings

Overview

Here we reanalyze a data presented in a recent tutorial on the use of MNLFA for traditional, level-based MI testing with multiple covariates (Godmann et al., 2026). The

⁵ Because this analysis involves 267 items, models are quite slow to converge, with both models used here taking 2+ hours to reach convergence in Mplus 9. We thus leave fitting further models to interested readers; the OSF contains data and scripts for the models we report here

Table 8*Fit comparison for no-DIF and DIF models of vocabulary acquisition of German children.*

Model	Parameters	Loglikelihood	LRT p-value	AIC	BIC
Mean and Variance Impact Only	536	-124310.454	–	249693	252417
With DIF Parameters	540	-124280.191	0.000	249640	252385

tutorial’s running example considers responses from 600 individuals to 14 items that aim to elicit two dimensions of paranormal beliefs: belief in extraordinary human abilities (e.g., dowsing, healing via laying of hands) and belief in unearthly beings, the latter of which is the focus of this example. As noted in the original tutorial, this data is a subset from a larger dataset cleaned for the purpose of creating a “clean” teaching dataset, so we use it mainly to demonstrate the flexibility of our approach with respect to the metric of the items, rather than to produce generalizable substantive insights. We follow the original researchers’ approach of treating the item responses on a five-point Likert scale as continuous (e.g., Rhemtulla et al., 2012; Robitzsch, 2020). We employ MLR standard errors. The original paper finds that on the Unearthly Beings construct, two items appear noninvariant with respect to their factor loading (and therefore should have their intercept modeled as noninvariant as well; Thissen, 2025, p. 18). Belief in devil possession is noninvariant with respect to both age and gender, discriminating better among men than among women, while belief in aliens is noninvariant with respect to gender only (with the slope again being higher for men).

Model Specification

Primarily for the purpose of illustrating the flexibility of the proposed modeling approach with respect to the metric of the indicators, we consider two models: one in which invariance is modeled as just described with no item groupings, and one in which the devil possession and alien belief items are treated as part of the same item group with respect to gender but *not* with respect to age. This demonstrates both the use of continuous indicators in a more traditional linear factor analysis setting (with moderation), as well as the fact that item groups do not need to be the same across covariates. Additionally, we free the residual variance of *all* items across both covariates via a log-linear specification (Bauer, 2017), in line with sequential MI testing that specifies the residuals as the last and most restrictive level of invariance, as well as a level that is not necessary for mean comparisons (Meredith, 1993; Sörbom, 1974). We elide the full model specification as it is quite repetitive with specifications already presented above, but we invite readers to review the full model specification in Mplus, run the models, and compare results using scripts and data found in the OSF.

Results

AIC, BIC, and a likelihood ratio test (with adjustments for χ^2 tests of differences with the MLR estimator) all favor the model in which both slope and intercept effects are applied at the group level with respect to gender instead of estimating them separately (readers are invited to run these models, as they are available in the OSF along with the data; we have also included Mplus out-files with results for readers who do not have access to Mplus). Table 9 contains parameter estimates for the DIF effects when estimated at the individual item level vs. treating the two DIF items (with respect to age) as part of a group. We do not over-interpret these results, both because the sample was a non-randomly selected example used in a teaching-oriented study not intended to be generalizable (Godmann et al., 2026), and because we are not aware of a clear theory on paranormal beliefs that would specifically group belief in aliens and devil possession as part

of a group of items with respect to gender DIF. Our primary point here is just that the approach we outline can be used in any context where MNLFA itself can be used, which includes continuous or categorical covariates as well as items that are binary, ordered categorical, nominal, continuous, or censored continuous (Bauer, 2017, 2023).

Table 9

Items from models of measurement noninvariance of beliefs in unearthly beings.

Item	Covariate	Parameter	Traditional DIF Model, Estimate (SE)	Model with Item Grouping by Gender, Estimate (SE)
Belief in Devil Possession	Baseline	Slope	0.520 (0.068)	0.541 (0.059)
	Male	Slope	0.233 (0.096)	0.197 (0.060)
	Age	Slope	0.097 (0.051)	0.099 (0.051)
	Baseline	Intercept	1.614 (0.057)	1.602 (0.053)
	Male	Intercept	0.299 (0.081)	0.318 (0.056)
	Age	Intercept	0.159 (0.040)	0.161 (0.040)
Belief in Aliens	Baseline	Slope	0.440 (0.058)	0.423 (0.052)
	Male	Slope	0.162 (0.092)	0.197 (0.060)
	Baseline	Intercept	1.603 (0.050)	1.613 (0.047)
	Male	Intercept	0.335 (0.081)	0.318 (0.056)

Discussion

Our aim in this paper has been to outline how MNLFA models can be specified to encode specific theories about item-level explanations for DIF and measurement noninvariance. Because MNLFA itself is so flexible with respect to the number and metric of the covariates as well as the metric of the indicators, we expect this approach to be

applicable in a wide variety of applied research settings. In line with recent calls to reconsider rote applications of DIF or MI analysis (Rohrer & Paulewicz, 2025; Sterner et al., 2026; Suk & Lyu, 2026), we provide researchers with a means to learn more from analyses of invariance. Of course, one departure of this framework from automated approaches to DIF analysis (e.g., Belzak & Bauer, 2024; Y. Chen et al., 2023) or sequential MI analysis (Meredith, 1993; Sörbom, 1974) is that it requires quite a bit more decision-making on the part of the researcher. Additionally, as is the case more generally with “explanatory” approaches that model item parameters (of any type) as functions of item properties, the resulting parameter estimates are not causal in nature—at best, we can hope to find a model *compatible* with prior theory, and to replicate similar findings in future studies with new datasets. This is not unique to the present study, but rather inherent to research that involves reasoning about unobservable entities using statistics.

Additionally, we note that this approach is not meant to substitute for, or discount the importance of, multidimensional approaches (e.g., Reckase, 2009), but we also note that this modeling approach is not isomorphic with a multidimensional approach. It is quite common to model mildly multidimensional data as unidimensional (Bonifay et al., 2015; Reckase et al., 1988; Reise et al., 2013) for a number of reasons. It may be the case that the dimensions are so highly correlated that they cannot be meaningfully distinguished statistically. It may be the case that the use of the scores requires a unidimensional interpretation, making a multidimensional solution practically unacceptable (AERA et al., 2014). It may be the case that the construct of interest can be plausibly formulated as a composite of sub-constructs or “strands” (Wilson, 2004, 2023) that hang together cross-sectionally, but may do so differently in different sub-populations. It may also be the case, especially in the context of MI testing, that one has evidence of unidimensional invariance up to a certain level (configural, metric) across the background variables (see Kolbe et al., 2024; Molenaar, 2021 for discussions of invariance in MNLFA), but no more, and that one is interested in testing item content-based theories about why

higher levels of invariance do not hold.

As a future direction, we note that the approach outlined in this paper is not computationally limited to item groupings. In principle one can define, and estimate a model with, multiple item covariates (for example, multiple non-exclusive item grouping variables, or fully continuous item covariates). The model constraint-based ML framework and the Bayesian framework employed in this study can both be extended straightforwardly to this case. However, we leave this as a future direction: many psychological instruments are relatively short, limiting the practical usefulness of a model with more than one item covariate to cases of very long instruments, or perhaps data from a large item bank (e.g., Kachergis et al., 2025). Our speculation is that an item grouping-based approach is the right level of granularity for many psychological measures in use today.

References

- Ackerman, T. A., & Ma, Y. (2024). Examining differential item functioning from a multidimensional IRT perspective. *Psychometrika*, 89(1), 4–41.
<https://doi.org/10.1007/s11336-024-09965-6>
- AERA, APA, & NCME. (2014). *Standards for Educational and Psychological Testing* [OCLC: ocn826867074]. American Educational Research Association.
- Andrich, D. (2004). Controversy and the Rasch model: A characteristic of incompatible paradigms? *Medical Care*, 42(1), 7.
<https://doi.org/10.1097/01.mlr.0000103528.48582.7c>
- Angoff, W. H. (1982). Use of difficulty and discrimination indices for detecting item bias. In R. A. Berk (Ed.). The Johns Hopkins Press.
- Angoff, W. H., & Ford, S. F. (1973). Item-race interaction on a test of scholastic aptitude. *Journal of Educational Measurement*, 10(95-106), 13.
<https://doi.org/10.1111/j.1745-3984.1973.tb00787.x>
- Bandalos, D. L. (2018). *Measurement Theory and Applications for the Social Sciences*. Guilford Press.
- Battaaz, M. (2020). Regularized Estimation of the Four-Parameter Logistic Model. *Psych*, 2(4), 269–278. <https://doi.org/10.3390/psych2040020>
- Battaaz, M. (2025). Differential item functioning detection across multiple groups. *British Journal of Mathematical and Statistical Psychology*, bmsp.70023.
<https://doi.org/10.1111/bmsp.70023>
- Bauer, D. J. (2017). A more general model for testing measurement invariance and differential item functioning. *Psychological Methods*, 22(3), 507–526.
<https://doi.org/10.1037/met0000077>
- Bauer, D. J. (2023). Enhancing measurement validity in diverse populations: Modern approaches to evaluating differential item functioning. *British Journal of*

- Mathematical and Statistical Psychology*, 76(3), 435–461.
<https://doi.org/10.1111/bmsp.12316>
- Bauer, D. J., Belzak, W. C. M., & Cole, V. T. (2020). Simplifying the assessment of measurement invariance over multiple background variables: Using regularized moderated nonlinear factor analysis to detect differential item functioning. *Structural Equation Modeling: A Multidisciplinary Journal*, 27(1), 43–55.
<https://doi.org/10.1080/10705511.2019.1642754>
- Bauer, D. J., & Hussong, A. M. (2009). Psychometric approaches for developing commensurate measures across independent studies: Traditional and new models. *Psychological Methods*, 14(2), 101–125. <https://doi.org/10.1037/a0015583>
- Belzak, W. C. M. (2023). The multidimensionality of measurement bias in high-stakes testing: Using machine learning to evaluate complex sources of differential item functioning. *Educational Measurement: Issues and Practice*, 42(1), 24–33.
<https://doi.org/10.1111/emip.12486>
- Belzak, W. C. M., & Bauer, D. J. (2020). Improving the assessment of measurement invariance: Using regularization to select anchor items and identify differential item functioning. *Psychological Methods*, 25(6), 673–690.
<https://doi.org/10.1037/met0000253>
- Belzak, W. C. M., & Bauer, D. J. (2024). Using regularization to identify measurement bias across multiple background characteristics: A penalized expectation–maximization algorithm. *Journal of Educational and Behavioral Statistics*, 49(6), 976–1012. <https://doi.org/10.3102/10769986231226439>
- Bentler, P. M. (1990). Comparative fit indexes in structural models. *Psychological Bulletin*, 107(2), 238–246. <https://doi.org/10.1037/0033-2909.107.2.238>
- Birnbaum, A. (1968). Some latent trait models and their use in inferring an examinee’s ability. In F. Lord & M. Novick (Eds.), *Statistical theories of mental test scores* (pp. 395–479). Addison-Wesley.

- Bohn, M., Prein, J., Engicht, J., Haun, D., Gagarina, N., & Koch, T. (2025). PREVIC: An adaptive parent report measure of expressive vocabulary in children between 3 and 8 years of age. *Behavior Research Methods*, 57(3), 95.
<https://doi.org/10.3758/s13428-025-02615-4>
- Bonifay, W. E., & Cai, L. (2017). On the complexity of item response theory models. *Multivariate Behavioral Research*, 52(4), 465–484.
<https://doi.org/10.1080/00273171.2017.1309262>
- Bonifay, W. E., Reise, S. P., Scheines, R., & Meijer, R. R. (2015). When Are multidimensional data unidimensional enough for structural equation modeling? An evaluation of the DETECT multidimensionality index. *Structural Equation Modeling: A Multidisciplinary Journal*, 22(4), 504–516.
<https://doi.org/10.1080/10705511.2014.938596>
- Braeken, J., Lehre, A.-C., & Marcq, K. (2026). Pearls and perils when interpreting item-specific results from international large-scale educational assessments: The case of the “Nordic” fractions. *Studies in Educational Evaluation*, 88, 101552.
<https://doi.org/10.1016/j.stueduc.2025.101552>
- Brandt, H., Chen, S. M., & Bauer, D. J. (2025). Bayesian penalty methods for evaluating measurement invariance in moderated nonlinear factor analysis. *Psychological Methods*, 30(3), 482–512. <https://doi.org/10.1037/met0000552>
- Burnham, K. P., & Anderson, D. R. (2004). Multimodel inference: Understanding AIC and BIC in model selection. *Sociological Methods & Research*, 33(2), 261–304.
<https://doi.org/10.1177/0049124104268644>
- Camilli, G. (1993). The case against item bias detection techniques based on internal criteria: Do item bias procedures obscure test fairness issues? In P. W. Holland & H. Wainer (Eds.), *Differential Item Functioning* (pp. 397–418). Routledge.
- Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., & Riddell, A. (2017). *Stan* : A probabilistic

- programming language. *Journal of Statistical Software*, 76(1).
<https://doi.org/10.18637/jss.v076.i01>
- Chen, S. M., Bauer, D. J., Belzak, W. M., & Brandt, H. (2022). Advantages of spike and slab priors for detecting differential item functioning relative to other Bayesian regularizing priors and frequentist lasso. *Structural Equation Modeling: A Multidisciplinary Journal*, 29(1), 122–139.
<https://doi.org/10.1080/10705511.2021.1948335>
- Chen, Y., Li, C., Ouyang, J., & Xu, G. (2023). DIF statistical inference without knowing anchor items. *Psychometrika*, 88(4), 1097–1122.
<https://doi.org/10.1007/s11336-023-09930-9>
- Cho, S.-J., Goodwin, A., Naveiras, M., & De Boeck, P. (2024). Modeling nonlinear effects of person-by-item covariates in explanatory item response models: Exploratory plots and modeling using smooth functions. *Journal of Educational Measurement*, 61(4), 595–623. <https://doi.org/10.1111/jedm.12410>
- Christensen, K. B. (2006). From Rasch scores to regression. *Journal of Applied Measurement*, 7(2), 184–191.
- Clifton, J. D. W. (2020). Managing validity versus reliability trade-offs in scale-building decisions. *Psychological Methods*, 25(3), 259–270.
<https://doi.org/10.1037/met0000236>
- Cortina, J. M., Sheng, Z., Keener, S. K., Keeler, K. R., Grubb, L. K., Schmitt, N., Tonidandel, S., Summerville, K. M., Heggstad, E. D., & Banks, G. C. (2020). From alpha to omega and beyond! A look at the past, present, and (possible) future of psychometric soundness in the Journal of Applied Psychology. *Journal of Applied Psychology*, 105(12), 1351–1381. <https://doi.org/10.1037/apl0000815>
- Curran, P. J., & Hussong, A. M. (2009). Integrative data analysis: The simultaneous analysis of multiple data sets. *Psychological Methods*, 14(2), 81–100.
<https://doi.org/10.1037/a0015914>

- De Boeck, P. (2008). Random item IRT models. *Psychometrika*, 73(4), 533–559.
<https://doi.org/10.1007/s11336-008-9092-x>
- De Boeck, P., & Wilson, M. (Eds.). (2004). *Explanatory item response models: A generalized linear and nonlinear approach*. Springer New York.
<https://doi.org/10.1007/978-1-4757-3990-9>
- Domingue, B. W., Braginsky, M., Caffrey-Maffei, L., Gilbert, J. B., Kanopka, K., Kapoor, R., Lee, H., Liu, Y., Nadela, S., Pan, G., Zhang, L., Zhang, S., & Frank, M. C. (2025). An introduction to the Item Response Warehouse (IRW): A resource for enhancing data usage in psychometrics. *Behavior Research Methods*, 57(10), 276. <https://doi.org/10.3758/s13428-025-02796-y>
- Embretson, S. E. (1999). Generating items during testing: Psychometric issues and models. *Psychometrika*, 64(4), 407–433. <https://doi.org/10.1007/BF02294564>
- Enders, C. K., Vera, J. D., Keller, B. T., Lenartowicz, A., & Loo, S. K. (2024). Building a simpler moderated nonlinear factor analysis model with Markov Chain Monte Carlo estimation. *Psychological Methods*. <https://doi.org/10.1037/met0000712>
- Fischer, G. H. (1973). The linear logistic test model as an instrument in educational research. *Acta Psychologica*, 37(6), 359–374.
[https://doi.org/10.1016/0001-6918\(73\)90003-6](https://doi.org/10.1016/0001-6918(73)90003-6)
- Flake, J. K., & Fried, E. I. (2020). Measurement schmeasurement: Questionable measurement practices and how to avoid them. *Advances in Methods and Practices in Psychological Science*, 3(4), 456–465. <https://doi.org/10.1177/251524592095239>
- Gagnon, S. G., & Nagle, R. J. (2000). Comparison of the revised and original versions of the Bayley Scales of Infant Development. *School Psychology International*, 21(3), 293–305. <https://doi.org/10.1177/0143034300213006>
- Godmann, H. R., Molenaar, D., Ziegler, M., & Pfadt, J. M. (2026). A tutorial on assessing measurement invariance with moderated (non-)linear factor analysis in JASP.
https://doi.org/10.31234/osf.io/6ftqg_v2

- Hallquist, M. N., & Wiley, J. F. (2018). *MplusAutomation* : An R package for facilitating large-scale latent variable analyses in M plus. *Structural Equation Modeling: A Multidisciplinary Journal*, 25(4), 621–638.
<https://doi.org/10.1080/10705511.2017.1402334>
- Halpin, P. F. (2024). Differential item functioning via robust scaling. *Psychometrika*, 89(3), 796–821. <https://doi.org/10.1007/s11336-024-09957-6>
- Halpin, P. F., & Gilbert, J. B. (2024, September). Testing whether reported treatment effects are unduly dependent on the specific outcome measure used [arXiv:2409.03502 [stat]]. <https://doi.org/10.48550/arXiv.2409.03502>
- Herngreen, W. P., Reerink, J. D., Van Noord-Zaadstra, B. M., Verloover-Vanhorick, S. P., & Ruys, J. H. (1992). SMOCC: Design of a representative cohort-study of live-born infants in the Netherlands. *The European Journal of Public Health*, 2(2), 117–122.
<https://doi.org/10.1093/eurpub/2.2.117>
- Holland, P. W., & Thayer, D. T. (1986). Differential item functioning and the Mantel-Haenszel procedure. *ETS Research Report Series*, 1986(2), 1–24.
<https://doi.org/10.1002/j.2330-8516.1986.tb00186.x>
- Hussey, I., Alsalti, T., Bosco, F., Elson, M., & Arslan, R. (2025). An aberrant abundance of Cronbach’s alpha values at .70. *Advances in Methods and Practices in Psychological Science*, 8(1), 25152459241287123. <https://doi.org/10.1177/25152459241287123>
- Jacobusse, G., Van Buuren, S., & Verkerk, P. H. (2006). An interval scale for development of children aged 0–2 years. *Statistics in Medicine*, 25(13), 2272–2283.
<https://doi.org/10.1002/sim.2351>
- Jöreskog, K. G. (1971). Simultaneous factor analysis in several populations. *Psychometrika*, 36(4), 409–426.
- Jöreskog, K. G., & Goldberger, A. S. (1975). Estimation of a model with multiple indicators and multiple causes of a single latent variable. *Journal of the American Statistical Association*, 70(351), 631. <https://doi.org/10.2307/2285946>

- Kachergis, G., O'Reilly, F., Braginsky, M., Xiao, X., Lightbody, A., Shannon, K. A., Watson, Z., Zhang, L., Zhu, R., Abutto, A. B., Ma, W. A., Long, B., Murray, T., Yeatman, J. D., Sulik, M. J., Obradovic, J., Jenkins, N., Ansari, D., Perfetti, M. C., ... Frank, M. C. (2025, December). Creation and validation of the LEVANTE core tasks: Internationalized measures of learning and development for children ages 5-12 years. https://doi.org/10.31234/osf.io/r4dhw_v1
- Kanopka, K., & Domingue, B. W. (2025). A position-sensitive mixture item response model. *Journal of Educational and Behavioral Statistics*, 50(6), 10769986241289399. <https://doi.org/10.3102/10769986241289399>
- Kolbe, L., Jorgensen, T. D., & Molenaar, D. (2021). The impact of unmodeled heteroskedasticity on assessing measurement invariance in single-group models. *Structural Equation Modeling: A Multidisciplinary Journal*, 28(1), 82–98. <https://doi.org/10.1080/10705511.2020.1766357>
- Kolbe, L., Molenaar, D., Jak, S., & Jorgensen, T. D. (2024). Assessing measurement invariance with moderated nonlinear factor analysis using the R package OpenMx. *Psychological Methods*, 29(2), 388–406. <https://doi.org/10.1037/met0000501>
- Lee, S., Han, S., & Choi, S. W. (2024). A Bayesian moderated nonlinear factor analysis approach for DIF detection under violation of the equal variance assumption. *Journal of Educational Measurement*, jedm.12388. <https://doi.org/10.1111/jedm.12388>
- Lee, S., Kim, S., & Choi, S. W. (2025). Anchor detection strategy in moderated non-linear factor analysis for differential item functioning (DIF). *Applied Psychological Measurement*, 01466216251401206. <https://doi.org/10.1177/01466216251401206>
- Lord, F. M. (1980). *Applications of Item Response Theory to Practical Testing Problems*. L. Erlbaum Associates.
- Maassen, E., D'Urso, E. D., Van Assen, M. A. L. M., Nuijten, M. B., De Roover, K., & Wicherts, J. M. (2025). The dire disregard of measurement invariance testing in

- psychological science. *Psychological Methods*, 30(5), 966–979.
<https://doi.org/10.1037/met0000624>
- Magis, D., Tuerlinckx, F., & De Boeck, P. (2015). Detection of differential item functioning using the lasso approach. *Journal of Educational and Behavioral Statistics*, 40(2), 111–135. <https://doi.org/10.3102/1076998614559747>
- Maris, G., & Bechger, T. (2009). On interpreting the model parameters for the three parameter logistic model. *Measurement: Interdisciplinary Research & Perspective*, 7(2), 75–88. <https://doi.org/10.1080/15366360903070385>
- Masters, G. N. (1982). A Rasch model for partial credit scoring. *Psychometrika*, 47(2), 149–174. <https://doi.org/10.1007/BF02296272>
- McCray, G., McCoy, D., Kariger, P., Janus, M., Black, M. M., Chang, S. M., Tofail, F., Eekhout, I., Waldman, M., Van Buuren, S., Khanam, R., Sazawal, S., Nizar, A., Schönbeck, Y., Zongo, A., Brentani, A., Zhang, Y., Dua, T., Cavallera, V., . . . Gladstone, M. (2023). The creation of the Global Scales for Early Development (GSED) for children aged 0–3 years: Combining subject matter expert judgements with big data. *BMJ Global Health*, 8(1), e009827.
<https://doi.org/10.1136/bmjgh-2022-009827>
- Meredith, W. (1964). Notes on factorial invariance. *Psychometrika*, 29(2), 177–185.
<https://doi.org/10.1007/BF02289699>
- Meredith, W. (1993). Measurement invariance, factor analysis and factorial invariance. *Psychometrika*, 58(4), 525–543. <https://doi.org/10.1007/BF02294825>
- Molenaar, D. (2021). A flexible moderated factor analysis approach to test for measurement invariance across a continuous variable. *Psychological Methods*, 26(6), 660–679. <https://doi.org/10.1037/met0000360>
- Muraki, E. (1992). A generalized partial credit model: Application of an EM algorithm. *ETS Research Report Series*. <https://doi.org/10.1002/j.2333-8504.1992.tb01436.x>

- Muthén, B. O. (1993). Goodness of fit with categorical and other non-normal variables. In K. A. Bollen & J. Long (Eds.), *Testing Structural Equation Models* (pp. 205–243). Sage.
- Muthén, B. O., & Lehman, J. (1985). Multiple group IRT modeling: Applications to item bias analysis. *Journal of Educational Statistics*, 10(2), 133–142.
<https://doi.org/10.3102/10769986010002133>
- Muthén, B. O., & Muthén, L. (2025). Mplus. <https://www.statmodel.com/index.shtml>
- Ouyang, J., Chen, Y., Li, C., & Xu, G. (2026). Accounting for measurement bias: A new framework for reliable country ranking in large-scale educational assessments. *Journal of the American Statistical Association*, 1–23.
<https://doi.org/10.1080/01621459.2026.2670732>
- Padgett, R. N. (2025). Multidimensional constructs and moderated linear and nonlinear factor analysis [Version Number: 2]. <https://doi.org/10.48550/ARXIV.2509.05443>
- Pan, T. (2018). Fitting the Rasch model under the logistic regression framework to reduce estimation bias. *Journal of Modern Applied Statistical Methods*, 17(1), jmasm.eP2645. <https://doi.org/10.22237/jmasm/1530028025>
- R Core Team. (2025). R: A language and environment for statistical computing.
<https://www.R-project.org/>
- Rasch, G. (1960). *Probabilistic Models for Some Intelligence and Attainment Tests*. Danish Institute for Educational Research.
- Reckase, M. D. (2009). *Multidimensional Item Response Theory*. Springer New York.
<https://doi.org/10.1007/978-0-387-89976-3>
- Reckase, M. D., Ackerman, T. A., & Carlson, J. E. (1988). Building a unidimensional test using multidimensional items. *Journal of Educational Measurement*, 25(3), 193–203.
<https://doi.org/10.1111/j.1745-3984.1988.tb00302.x>

- Reise, S. P., Bonifay, W. E., & Haviland, M. G. (2013). Scoring and Modeling Psychological Measures in the Presence of Multidimensionality. *Journal of Personality Assessment, 95*(2), 129–140. <https://doi.org/10.1080/00223891.2012.725437>
- Rhemtulla, M., Brosseau-Liard, P. É., & Savalei, V. (2012). When can categorical variables be treated as continuous? A comparison of robust continuous and categorical SEM estimation methods under suboptimal conditions. *Psychological Methods, 17*(3), 354–373. <https://doi.org/10.1037/a0029315>
- Robitzsch, A. (2020). Why ordinal variables can (almost) always be treated as continuous variables: Clarifying assumptions of robust continuous and ordinal factor analysis estimation methods. *Frontiers in Education, 5*, 589965. <https://doi.org/10.3389/feduc.2020.589965>
- Robitzsch, A. (2024). Smooth information criterion for regularized estimation of item response models. *Algorithms, 17*(4), 153. <https://doi.org/10.3390/a17040153>
- Robitzsch, A., & Lüdtke, O. (2023). Why full, partial, or approximate measurement invariance are not a prerequisite for meaningful and valid group comparisons. *Structural Equation Modeling: A Multidisciplinary Journal, 30*(6), 859–870. <https://doi.org/10.1080/10705511.2023.2191292>
- Rohrer, J. M., & Paulewicz, B. (2025). Rethinking measurement invariance causally. *Current Research in Ecological and Social Psychology, 9*, 100241. <https://doi.org/10.1016/j.cresp.2025.100241>
- Samejima, F. (1969). Estimation of latent ability using a response pattern of graded scores. *Psychometrika Monograph Supplement, 34*(2, pt. 2), 100.
- Satorra, A., & Bentler, P. M. (2010). Ensuring positiveness of the scaled difference chi-square test statistic. *Psychometrika, 75*(2), 243–248. <https://doi.org/10.1007/s11336-009-9135-y>

- Schauberg, G., & Mair, P. (2020). A regularization approach for the detection of differential item functioning in generalized partial credit models. *Behavior Research Methods*, 52(1), 279–294. <https://doi.org/10.3758/s13428-019-01224-2>
- Sörbom, D. (1974). A general method for studying differences in factor means and factor structure between groups. *British Journal of Mathematical and Statistical Psychology*, 27(2), 229–239. <https://doi.org/10.1111/j.2044-8317.1974.tb00543.x>
- Stan Development Team. (2025). RStan: The R interface to Stan. mc-stan.org
- Sterner, P., Pargent, F., Deffner, D., & Goretzko, D. (2024). A causal framework for the comparability of latent variables. *Structural Equation Modeling: A Multidisciplinary Journal*, 31(5), 747–758. <https://doi.org/10.1080/10705511.2024.2339396>
- Sterner, P., Pargent, F., & Goretzko, D. (2026). Don't let MI be misunderstood: A causal view on measurement invariance. *Current Research in Ecological and Social Psychology*, 10, 100261. <https://doi.org/10.1016/j.cresp.2025.100261>
- Student, S. R. (2026). Causal parameter moderation: Applying moderated nonlinear factor analysis to causal inference with latent outcomes. *Journal of Educational and Behavioral Statistics*, Advance online publication. <https://doi.org/10.3102/10769986251414869>
- Student, S. R., & McCormick, E. M. (2025). Digital module 38: Differential item functioning by multiple variables using moderated nonlinear factor analysis. *Educational Measurement: Issues and Practice*, 44(2), 39–41. <https://doi.org/10.1111/emip.12669>
- Suk, Y., & Lyu, W. (2026). Rethinking item fairness using single world intervention graphs. *Journal of Educational and Behavioral Statistics*, 10769986251415570. <https://doi.org/10.3102/10769986251415570>
- Swaminathan, H., & Rogers, H. J. (1990). Detecting differential item functioning using logistic regression procedures. *Journal of Educational Measurement*, 27(4), 361–370. <https://doi.org/10.1111/j.1745-3984.1990.tb00754.x>

- Tay, L., Newman, D. A., & Vermunt, J. K. (2011). Using mixed-measurement item response theory with covariates (MM-IRT-C) to ascertain observed and unobserved measurement equivalence. *Organizational Research Methods*, 14(1), 147–176.
<https://doi.org/10.1177/1094428110366037>
- Tay, L., Vermunt, J. K., & Wang, C. (2013). Assessing the item response theory with covariate (IRT-C) procedure for ascertaining differential item functioning. *International Journal of Testing*, 13(3), 201–222.
<https://doi.org/10.1080/15305058.2012.692415>
- Thissen, D. (2025). A review of some of the history of factorial invariance and differential item functioning. *Multivariate Behavioral Research*, 60(2), 211–235.
<https://doi.org/10.1080/00273171.2024.2396148>
- Tutz, G., & Schauberger, G. (2015). A penalty approach to differential item functioning in Rasch models. *Psychometrika*, 80(1), 21–43.
<https://doi.org/10.1007/s11336-013-9377-6>
- Wang, C., Zhu, R., & Xu, G. (2023). Using lasso and adaptive lasso to identify DIF in multidimensional 2PL models. *Multivariate Behavioral Research*, 58(2), 387–407.
<https://doi.org/10.1080/00273171.2021.1985950>
- Weber, A. M., Rubio-Codina, M., Walker, S. P., Van Buuren, S., Eekhout, I., Grantham-McGregor, S. M., Araujo, M. C., Chang, S. M., Fernald, L. C., Hamadani, J. D., Hanlon, C., Karam, S. M., Lozoff, B., Ratsifandrihamanana, L., Richter, L., & Black, M. M. (2019). The D-score: A metric for interpreting the early development of infants and toddlers across global settings. *BMJ Global Health*, 4(6), e001724. <https://doi.org/10.1136/bmjgh-2019-001724>
- Wilson, M. (2004). *Constructing measures: An item response modeling approach*. Lawrence Erlbaum Associates.
- Wilson, M. (2023). *Constructing measures: An item response modeling approach* (2nd ed.). Routledge.

- Woods, C. M., & Grimm, K. J. (2011). Testing for nonuniform differential item functioning with Multiple Indicator Multiple Cause models. *Applied Psychological Measurement*, 35(5), 339–361. <https://doi.org/10.1177/0146621611405984>
- Zhang, L., Liu, Y., Molenaar, D., Gilbert, J., Kanopka, K., & Domingue, B. W. (2025, December). Realistic Simulation of Item Difficulties. https://doi.org/10.31234/osf.io/jbhxy_v4

Appendix A

2PL Parameter Estimates from Focal Analysis

Table A1

Relevant parameter estimates for moderated 2PL models for child development.

Model	Parameter	Interpretation	Estimate (SE)
Latent Regression	α_1	Difference in mean development for female vs. male children	0.107 (0.093)
Item Group Moderated Intercepts	α_1	Difference in mean development for female vs. male children <i>as represented by cognitive items (anchor item group)</i>	0.430 (0.122)
	ξ_{11}	Intercept effect on fine motor items	-0.373 (0.169)
	ξ_{21}	Intercept effect on gross motor items	-0.954 (0.215)

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Table A1 (continued)

Model	Parameter	Interpretation	Estimate (SE)
Item Group Moderated Intercepts and Slopes	α_1	Difference in mean development for female vs. male children <i>as represented by cognitive items (anchor item group)</i>	0.430 (0.122)
	ξ_{11}	Intercept effect on fine motor items	-0.415 (0.172)
	ξ_{21}	Intercept effect on gross motor items	-0.767 (0.242)
	κ_{11}	Slope effect on fine motor items	-0.161 (0.223)
	κ_{21}	Slope effect on gross motor items	0.531 (0.256)

Appendix B

Full-data Replication of Focal Analysis

Here we present results from re-estimating the focal analysis models on the full 1723-person dataset instead of the complete-data subset. The larger sample predictably leads to smaller standard errors, but results are otherwise qualitatively unchanged.

Table B1

Fit comparison of moderated Rasch models for child development with full sample.

Model	Loglikelihood	Parameters	AIC	BIC	LRT p-value
Latent Regression	-6089.66	14	12207	12284	—
Item Group Moderated	-6066.33	16	12165	12252	0.000
Full DIF	-6063.07	22	12170	12290	0.369

Note. AIC = Akaike Information Criterion, BIC = Bayesian Information Criterion, LRT p-value = p-value based on likelihood ratio test of difference in model fit via χ^2 distribution with degrees of freedom equal to difference in number of parameters for adjacent models. **Latent Regression** = standard Rasch model with regression on sex; **Item Group Moderated** = Rasch model with one parameter for intercept DIF across all fine motor items and one parameter for intercept DIF across all gross motor items; **Full DIF** = Rasch model with DIF separately estimated for each gross and fine motor item.

Table B2

Relevant parameter estimates for moderated Rasch models for child development with full sample.

Model		Parameter	Interpretation	Estimate (SE)	Standardized (SE)
Latent Regression	Regres-	α_1	Difference in mean development for female vs. male children	0.419 (0.094)	0.271 (0.060)
Item Moderated	Group	α_1	Difference in mean development for female vs. male children <i>as</i> <i>represented by</i> <i>cognitive items</i> <i>(anchor item</i> <i>group)</i>	0.795 (0.110)	0.500 (0.066)
		ξ_{11}	Intercept effect on fine motor items	-0.595 (0.122)	-0.595 (0.122)
		ξ_{21}	Intercept effect on gross motor items	-0.776 (0.123)	-0.776 (0.123)

Table B3

Fit comparison of moderated 2PL models for child development with full sample.

Model	Loglikelihood	Parameters	AIC	BIC	LRT p-value
Latent Regression	-6045.49	25	12141	12277	—
Item Group Moderated Intercepts	-6023.24	27	12100	12247	0.000
Item Group Moderated Intercepts and Slopes	-6021.06	29	12100	12258	0.113

Note. AIC = Akaike Information Criterion, BIC = Bayesian Information Criterion, LRT p-value = p-value based on likelihood ratio test of difference in model fit via χ^2 distribution with two degrees of freedom, comparing adjacent models (i.e., moderated intercepts vs. unmoderated, moderated intercepts and slopes vs. moderated intercepts only).

Table B4*Relevant parameter estimates for moderated 2PL models for child development.*

Model	Parameter	Interpretation	Estimate (SE)
Latent Regression	α_1	Difference in mean development for female vs. male children	0.107 (0.093)
Item Group Moderated Intercepts	α_1	Difference in mean development for female vs. male children <i>as represented by cognitive items (anchor item group)</i>	0.430 (0.122)
	ξ_{11}	Intercept effect on fine motor items	-0.373 (0.169)
	ξ_{21}	Intercept effect on gross motor items	-0.954 (0.215)
Item Group Moderated Intercepts and Slopes	α_1	Difference in mean development for female vs. male children <i>as represented by cognitive items (anchor item group)</i>	0.430 (0.122)

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Table B4 (continued)

Model	Parameter	Interpretation	Estimate (SE)
	ξ_{11}	Intercept effect on fine motor items	-0.415 (0.172)
	ξ_{21}	Intercept effect on gross motor items	-0.767 (0.242)
	κ_{11}	Slope effect on fine motor items	-0.161 (0.223)
	κ_{21}	Slope effect on gross motor items	0.531 (0.256)

Appendix C

Simulation Results for Baseline Item Parameters

As noted in the main text recovery of baseline item parameters in the simulation study is excellent in all conditions. Below we summarize these results separately for item intercepts and slopes.

Table C1

Summary Simulation Results for Baseline Item Parameters

SS	Parameter	Bias	Coverage	SE Bias
250	Intercept	-0.07-0.06	0.90-0.99	-0.05-0.03
250	Slope	-0.03-0.07	0.90-1.00	-0.04-0.03
500	Intercept	-0.04-0.04	0.91-0.99	-0.02-0.02
500	Slope	-0.02-0.05	0.90-0.99	-0.02-0.03
1000	Intercept	-0.02-0.03	0.89-0.99	-0.02-0.01
1000	Slope	-0.02-0.02	0.90-0.99	-0.02-0.02
2000	Intercept	-0.02-0.02	0.90-1.00	-0.01-0.01
2000	Slope	-0.01-0.02	0.91-0.99	-0.01-0.01